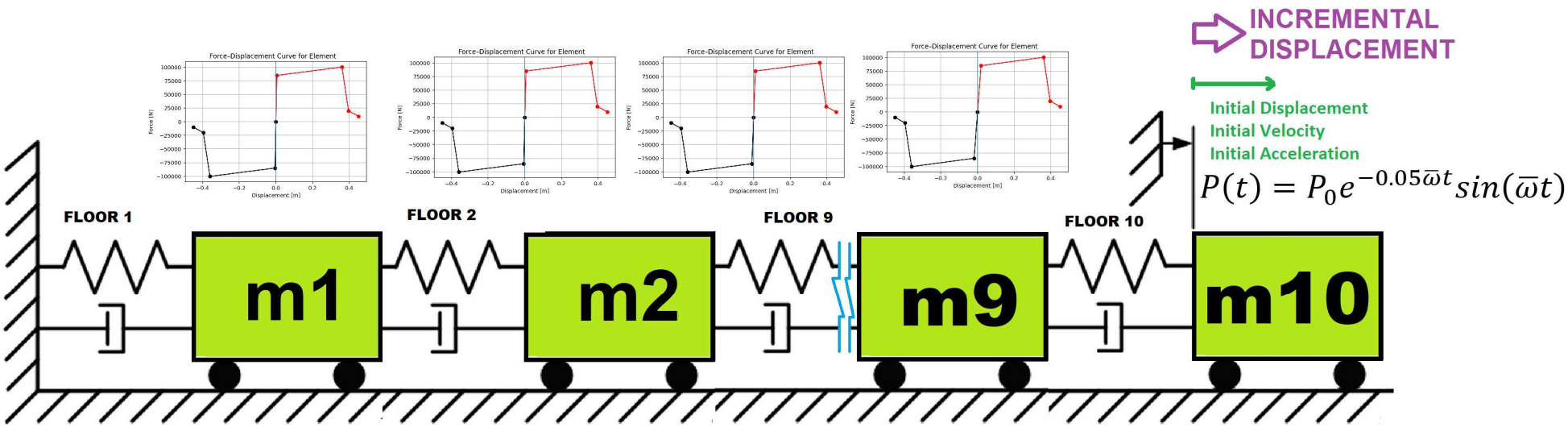
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

SENSITIVITY ANALYSIS OF COLUMN YIELD STRENGTH, ELASTIC STIFFNESS, DUCTILITY RATIO, OVER-STRENGTH FACTOR AND OPENSEES VIA PUSHOVER ANALYSIS OF A MULTI-DEGREE-OF-FREEDOM STRUCTURE AND EVALUATION OF A MULTILINEAR FITTING CURVE FOR STRUCTURAL ELASTIC STIFFNESS, PLASTIC STIFFNESS, DUCTILITY RATIO, AND OVER-STRENGTH FACTOR

THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$Structural\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

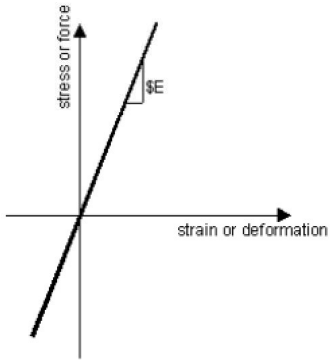
Δ_d = *Lateral Displaement from Dynamic Analysis*

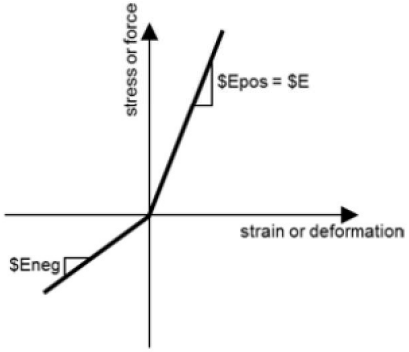
Δ_y = *Lateral Yield Displaement from Pushover Analysis*

Δ_u = *Lateral Ultimate Displaement from Pushover Analysis*

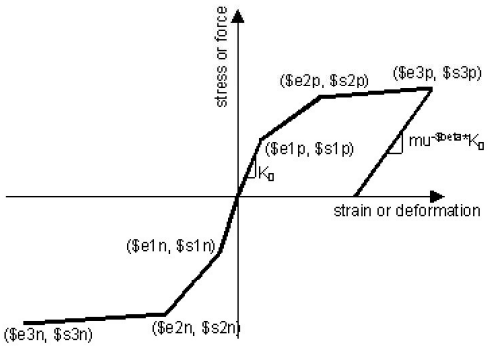
$$P(t) = P_0 e^{-0.05\bar{\omega}t} sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

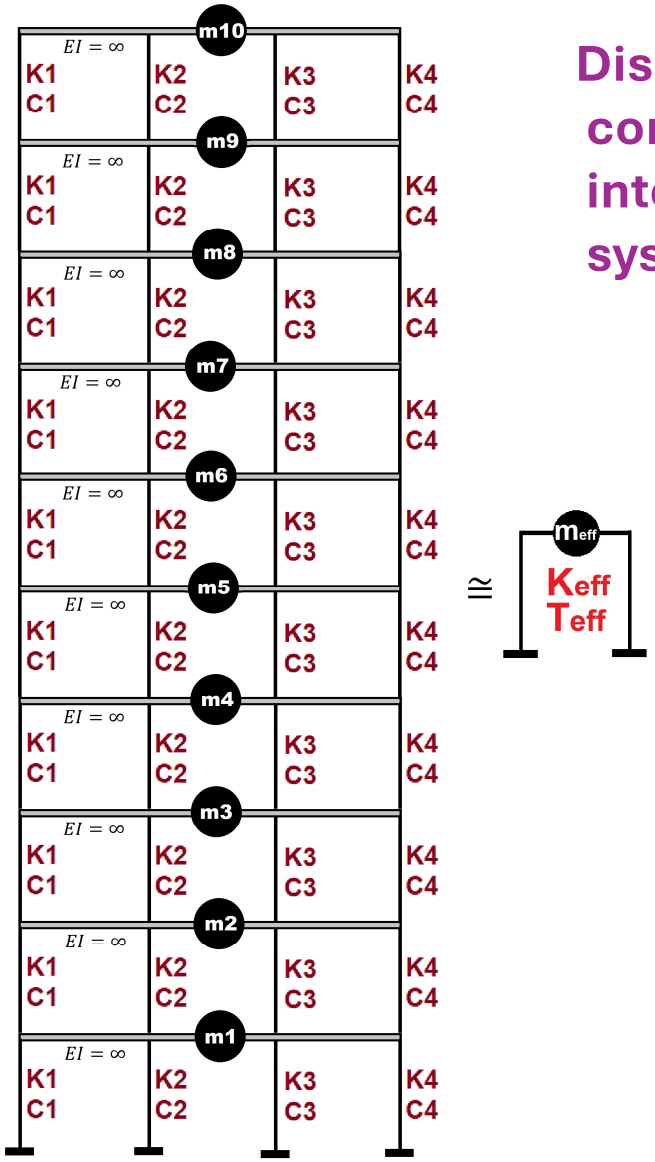




ELASTIC MATERIAL



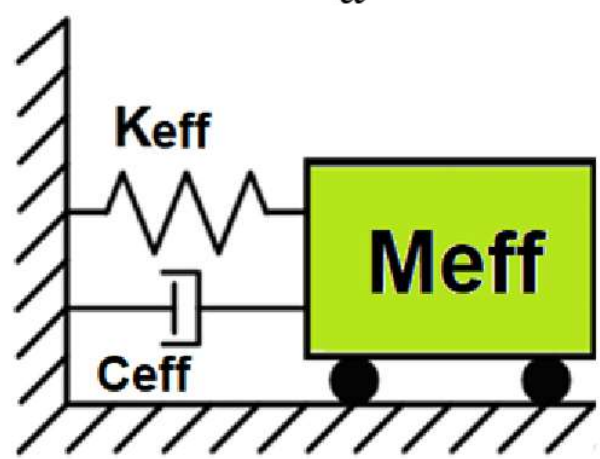
INELASTIC MATERIAL



Displacement-based seismic analysis transformation, converting a multi-degree-of-freedom (MDOF) system into an equivalent single-degree-of-freedom (SDOF) system for seismic assessment

$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i} \qquad m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d} \qquad T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$



Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

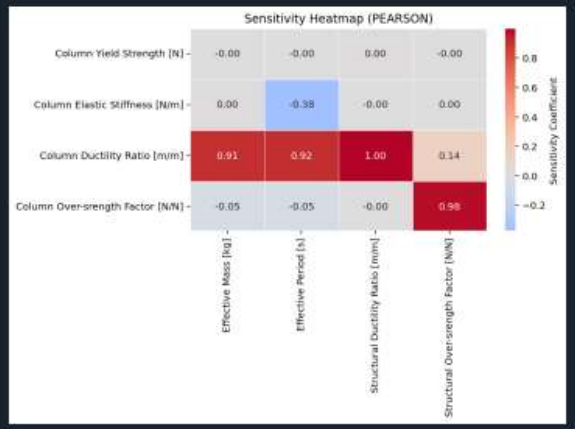
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MDOF+\FY_KE_DUCT_OSF_SENSITIVITY_PUSHOVER_MDOF.py

FY_KE_DUCT_OSF_SEN...Y_PUSHOVER_MDOF.py

```
##########################################################################################
#                    >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
#          SENSITIVITY ANALYSIS OF COLUMN YIELD STRENGTH, ELASTIC STIFFNESS, DUCTILITY RATIO,
#    OVER-STRENGTH FACTOR AND OPENSEES VIA PUSHOVER ANALYSIS OF A MULTI-DEGREE-OF-FREEDOM STRUCTURE
#           AND EVALUATION OF A MULTILINEAR FITTING CURVE FOR STRUCTURAL ELASTIC STIFFNESS,
#                     PLASTIC STIFFNESS, DUCTILITY RATIO, AND OVER-STRENGTH FACTOR
#------------------------------------------------------------------------------------------
# EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANALY
#------------------------------------------------------------------------------------------
#                  THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
#                                      EMAIL: salar.d.ghashghaei@gmail.com
##########################################################################################
"""
# ==========================================================================================
#   SENSITIVITY ANALYSIS OF A MULTI-DEGREE-OF-FREEDOM (MDOF) BUILDING FRAME
#   ------------------------------------------------------------------------------
#   Effect of Column Ductility Ratio on the Equivalent SDOF System and on the
#   Multilinear Idealization of the Pushover Curve
#
#   METHODOLOGY
#   -----------
#   1.  A 10-story MDOF shear-building frame is modelled in OpenSeesPy
#       (10 lumped masses, 4 parallel zeroLength springs per floor).
#
#   2.  The COLUMN DUCTILITY RATIO is chosen as the single design parameter and is
#       swept over 21 equally spaced values between DUCT_MIN and DUCT_MAX.
#
#   3.  For every value of the swept parameter, a nonlinear PUSHOVER ANALYSIS is
#       carried out under displacement control applied at the top node.
#
#   4.  The resulting base-shear / displacement curve is post-processed to derive the
#       EQUIVALENT SDOF SYSTEM (effective displacement, mass, stiffness, and period)
#       for displacement-based seismic design.
#
```

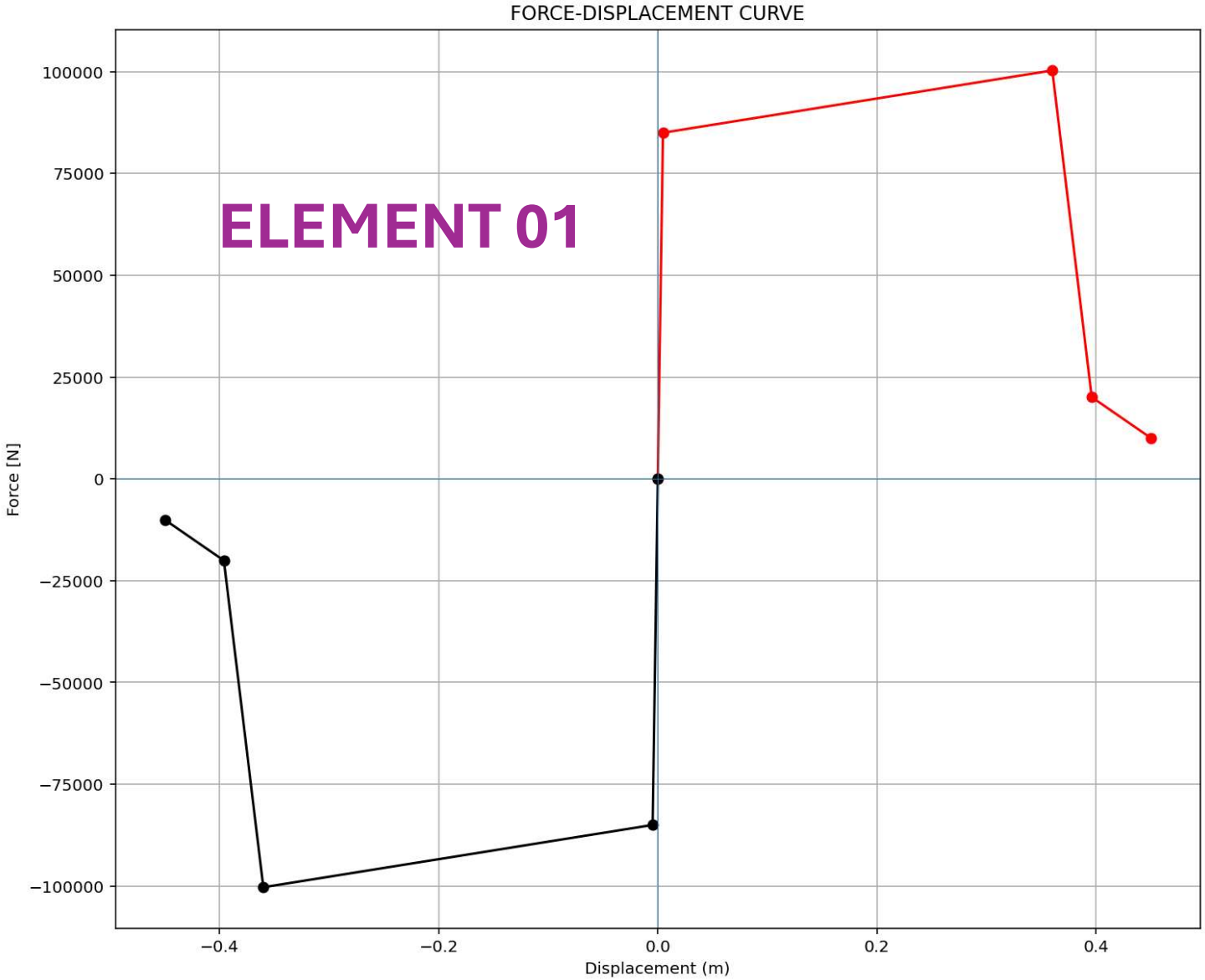
...4_COLUMNS_8_ANA\FY_KE_DUCT_OSF_SENSITIVITY_PUSHOVER_MDOF+

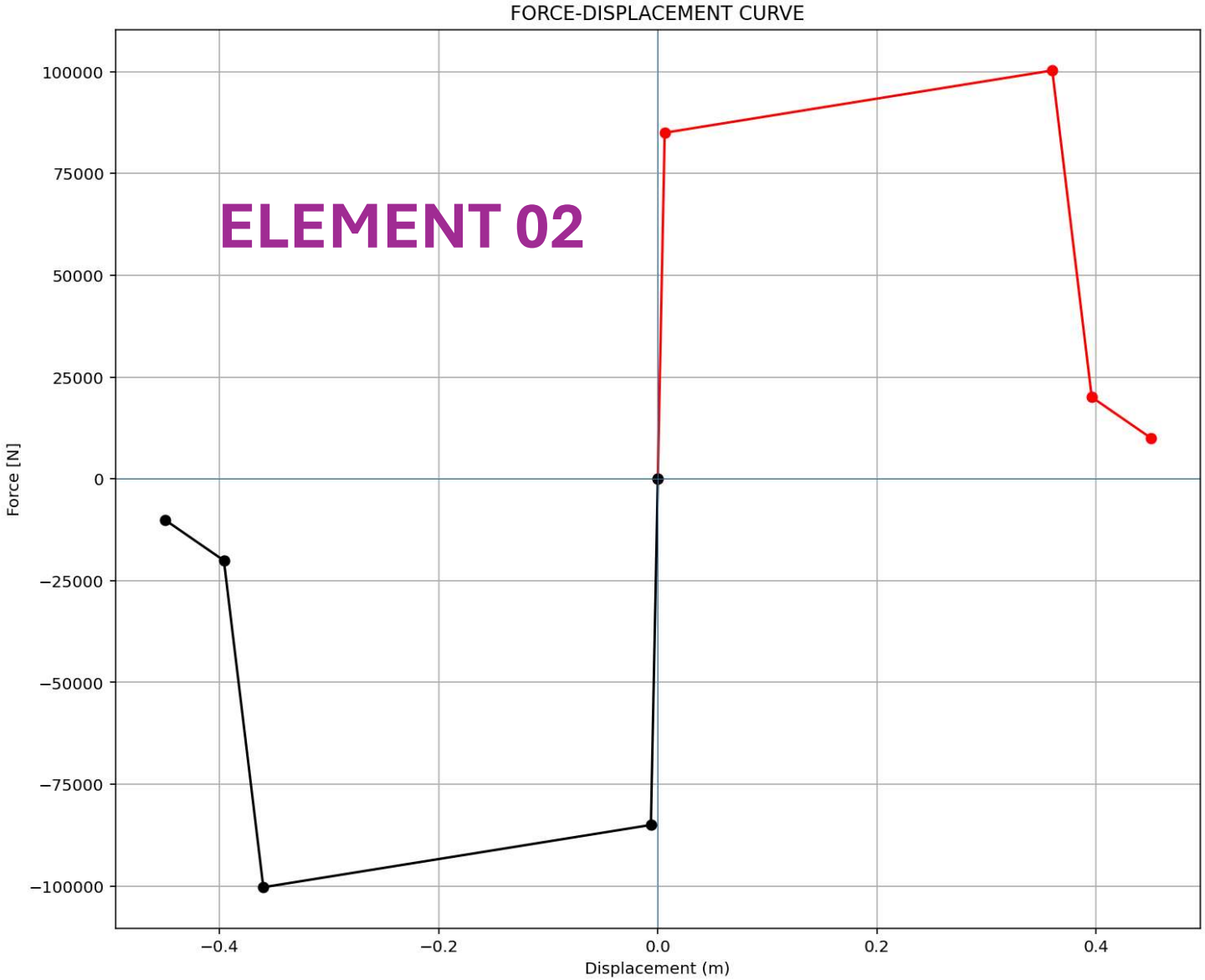
32 %

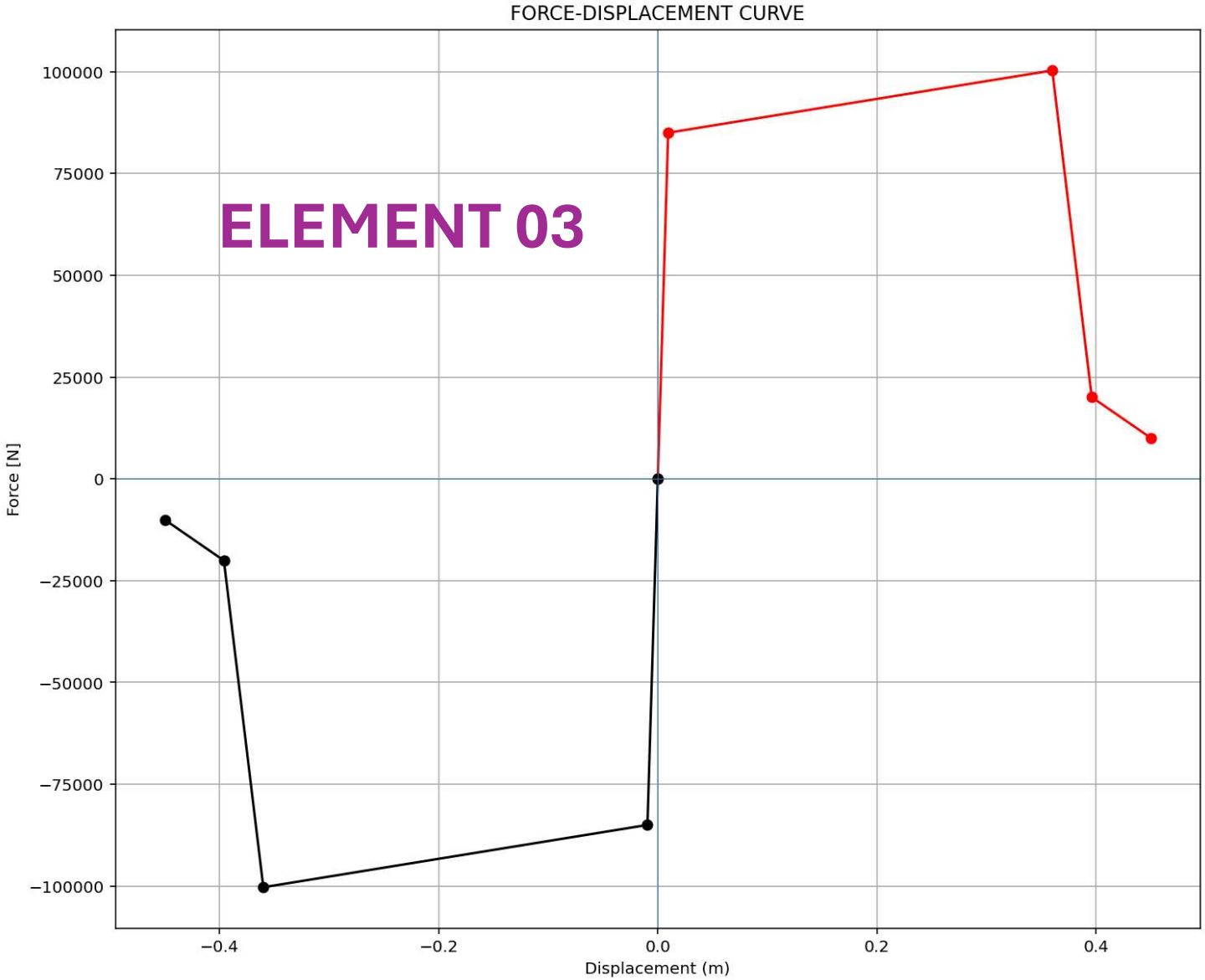


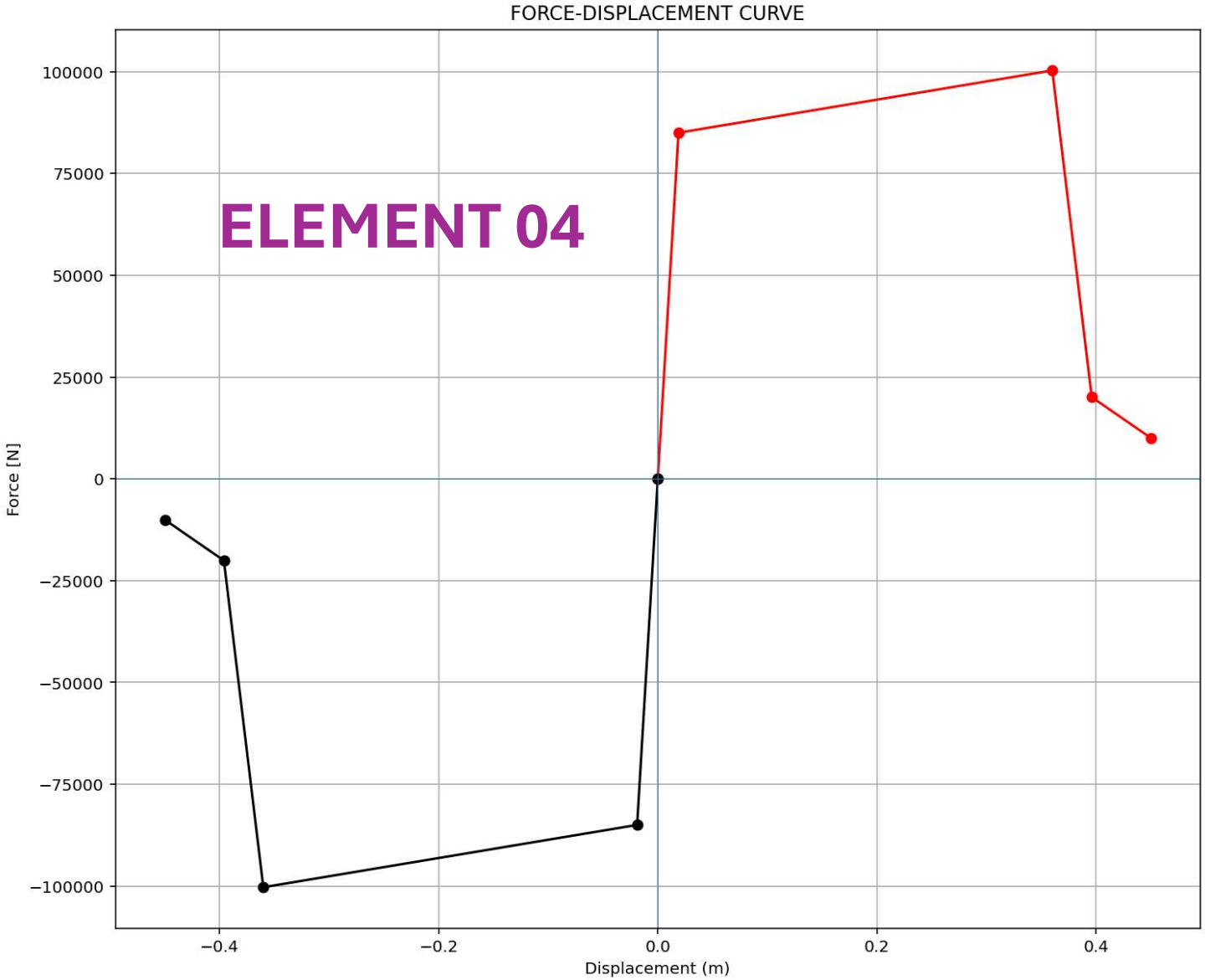
IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 1652, Col 82 | UTF-8 | CRLF | RW | Mem 75%


FORCE-DISPLACEMENT CURVE
ELEMENT 01
Force [N]
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.4
−0.2
0.0
0.2
0.4


FORCE-DISPLACEMENT CURVE
ELEMENT 02
Force [N]
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.4
−0.2
0.0
0.2
0.4


FORCE-DISPLACEMENT CURVE
ELEMENT 03
Force [N]
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.4
−0.2
0.0
0.2
0.4


FORCE-DISPLACEMENT CURVE
ELEMENT 04
Force [N]
Displacement (m)
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
−0.4
−0.2
0.0
0.2
0.4

PUSHOVER ANALYSIS

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

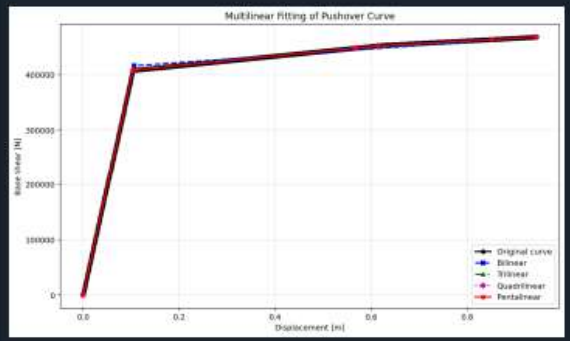
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...MDOF+\FY_KE_DUCT_OSF_SENSITIVITY_PUSHOVER_MDOF.py

FY_KE_DUCT_OSF_SEN...Y_PUSHOVER_MDOF.py

```
##########################################################################################
#                    >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
#           SENSITIVITY ANALYSIS OF COLUMN YIELD STRENGTH, ELASTIC STIFFNESS, DUCTILITY RATIO,
#     OVER-STRENGTH FACTOR AND OPENSEES VIA PUSHOVER ANALYSIS OF A MULTI-DEGREE-OF-FREEDOM STRUCTURE
#            AND EVALUATION OF A MULTILINEAR FITTING CURVE FOR STRUCTURAL ELASTIC STIFFNESS,
#                     PLASTIC STIFFNESS, DUCTILITY RATIO, AND OVER-STRENGTH FACTOR
#------------------------------------------------------------------------------------------
# EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANALY
#------------------------------------------------------------------------------------------
#                  THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
#                                  EMAIL: salar.d.ghashghaei@gmail.com
##########################################################################################
"""
# ==============================================================================
#   SENSITIVITY ANALYSIS OF A MULTI-DEGREE-OF-FREEDOM (MDOF) BUILDING FRAME
#   ----------------------------------------------------------------------------
#   Effect of Column Yield Strength, Elastic Stiffness, Ductility Ratio, Over-stength Fctor, on the Equ
#   Multilinear Idealization of the Pushover Curve
#
#   METHODOLOGY
#   -----------
#   1.  A 10-story MDOF shear-building frame is modelled in OpenSeesPy
#       (10 lumped masses, 4 parallel zeroLength springs per floor).
#
#   2.  The COLUMN DUCTILITY RATIO is chosen as the single design parameter and is
#       swept over 21 equally spaced values between DUCT_MIN and DUCT_MAX.
#
#   3.  For every value of the swept parameter, a nonlinear PUSHOVER ANALYSIS is
#       carried out under displacement control applied at the top node.
#
#   4.  The resulting base-shear / displacement curve is post-processed to derive the
#       EQUIVALENT SDOF SYSTEM (effective displacement, mass, stiffness, and period)
#       for displacement-based seismic design.
#
```

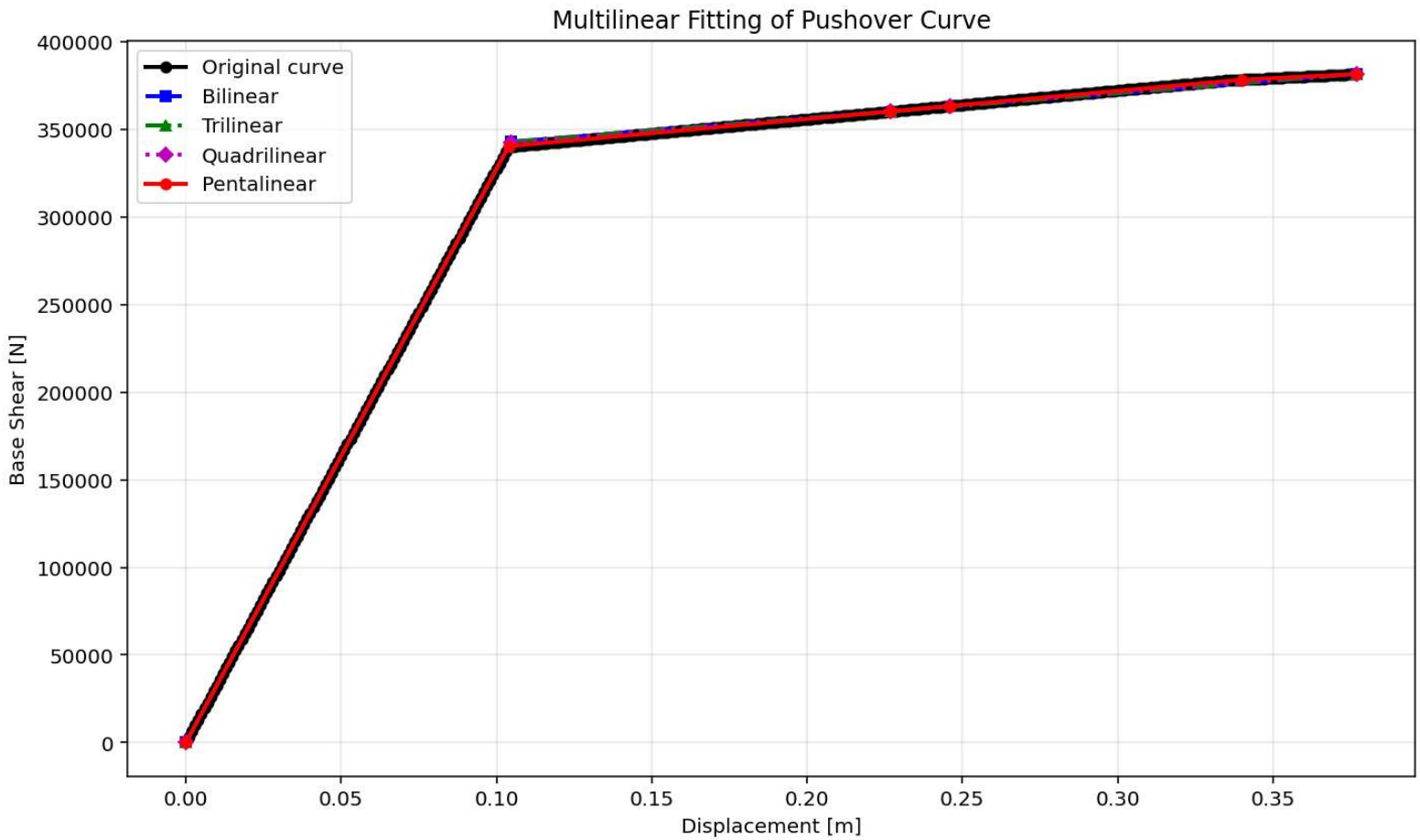
...4_COLUMNS_8_ANA\FY_KE_DUCT_OSF_SENSITIVITY_PUSHOVER_MDOF+

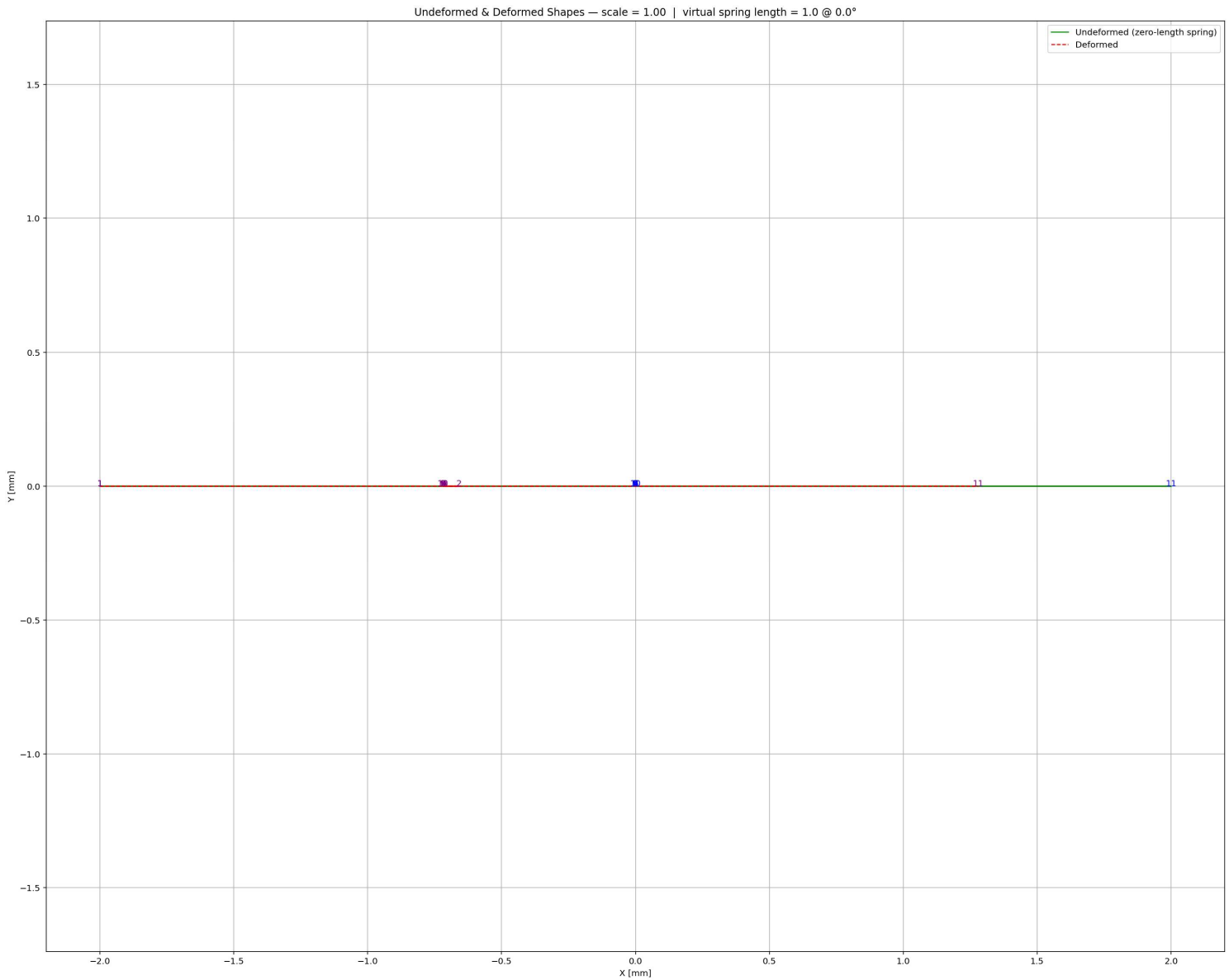
25 %



IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

Save file | Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 17, Col 47 | UTF-8 | CRLF | RW | Mem 68%


Multilinear Fitting of Pushover Curve
Original curve
Bilinear
Trilinear
Quadrilinear
Pentalinear
Base Shear [N]
400000
350000
300000
250000
200000
150000
100000
50000
0
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
Displacement [m]


Undeformed & Deformed Shapes — scale = 1.00 | virtual spring length = 1.0 @ 0.0°
Undeformed (zero-length spring)
Deformed
Y [mm]
X [mm]
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
1
2
11
11

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...DOF+\COMPUTE_EFFECTIVE_PROPERTIES_FUN_PUSHOVER.py

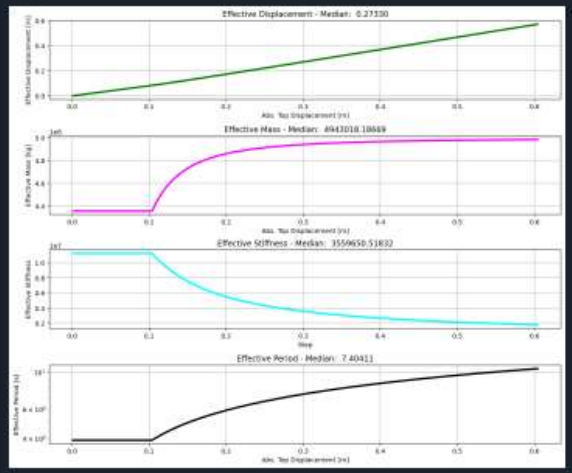
FY_KE_DUCT_OSF_SEN...Y_PUSHOVER_MDOF.py | COMPUTE_EFFECTIVE_...ES_FUN_PUSHOVER.py

```python
#%% EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED SEISMIC DESIGN PROCEDURE WITH PUSHOVER ANAI
# Change MDOF to SDOF System with Displacement Based Design Concept
# THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
"""
This script implements a displacement-based pushover transformation,
 converting a multi-degree-of-freedom (MDOF) system into an equivalent
 single-degree-of-freedom (SDOF) system for seismic assessment.
 It calculates effective modal properties—displacement, mass, and
 stiffness—by weighting element forces and nodal displacements according
 to a presumed deformed shape.
 The derived effective period provides a simplified dynamic characteristic
 for performance-based engineering. The visualizations effectively track the
 evolution of these equivalent parameters throughout the nonlinear analysis steps.
"""
#-----------------------------------------------------------------------
displacement_X_1 = np.array(list(node_displacements.values())[1])   # DOF 02
displacement_X_2 = np.array(list(node_displacements.values())[2])   # DOF 03
displacement_X_3 = np.array(list(node_displacements.values())[3])   # DOF 04
displacement_X_4 = np.array(list(node_displacements.values())[4])   # DOF 05
displacement_X_5 = np.array(list(node_displacements.values())[5])   # DOF 06
displacement_X_6 = np.array(list(node_displacements.values())[6])   # DOF 07
displacement_X_7 = np.array(list(node_displacements.values())[7])   # DOF 08
displacement_X_8 = np.array(list(node_displacements.values())[8])   # DOF 09
displacement_X_9 = np.array(list(node_displacements.values())[9])   # DOF 10
displacement_X_10 = np.array(list(node_displacements.values())[10])   # DOF 11

ele_force_01 = np.array(list(ele_force.values())[0])   # ELEMENT 01
ele_force_02 = np.array(list(ele_force.values())[1])   # ELEMENT 02
ele_force_03 = np.array(list(ele_force.values())[2])   # ELEMENT 03
ele_force_04 = np.array(list(ele_force.values())[3])   # ELEMENT 04
ele_force_05 = np.array(list(ele_force.values())[4])   # ELEMENT 05
ele_force_06 = np.array(list(ele_force.values())[5])   # ELEMENT 06
ele_force_07 = np.array(list(ele_force.values())[6])   # ELEMENT 07
ele_force_08 = np.array(list(ele_force.values())[7])   # ELEMENT 08
```

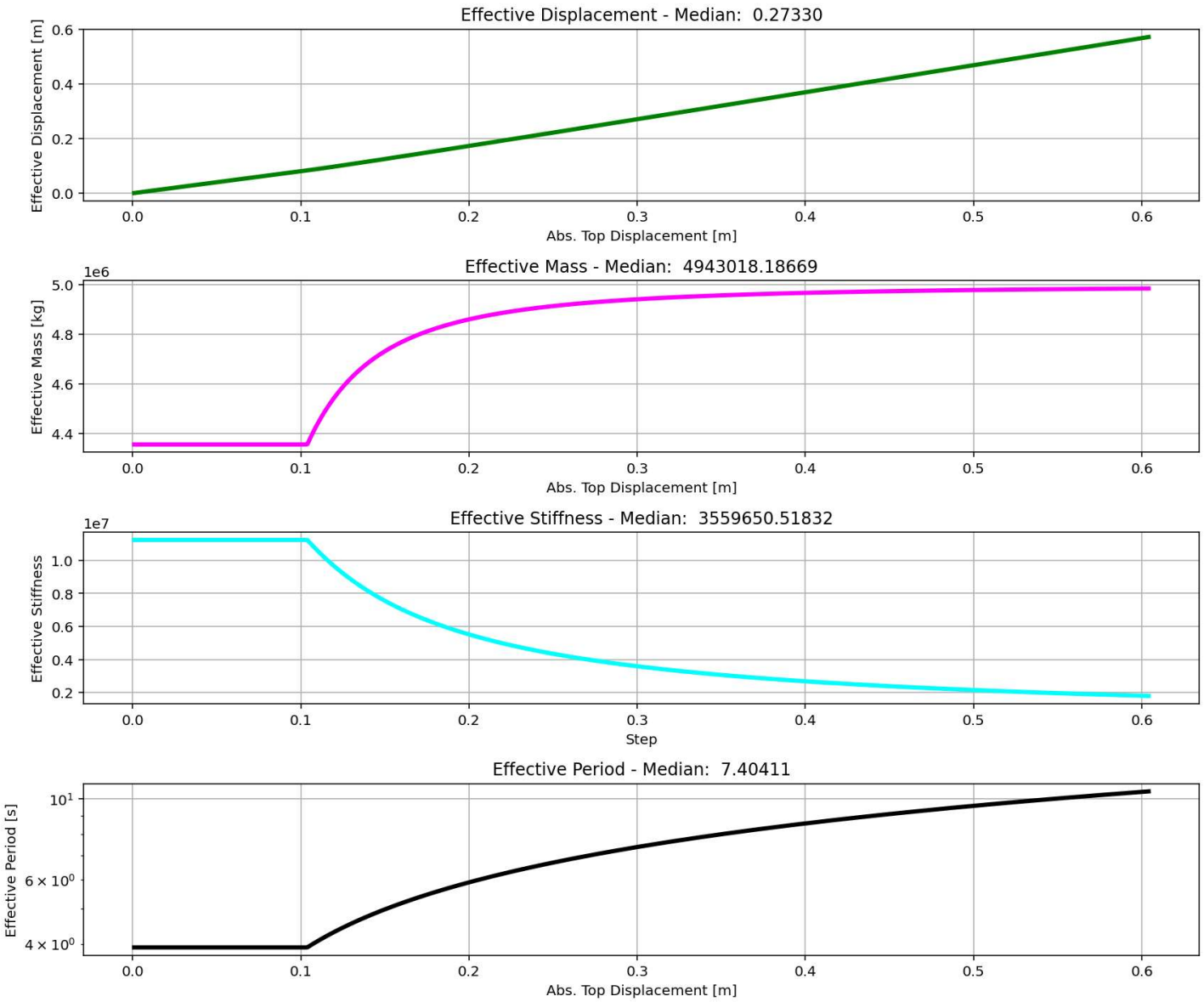
...4_COLUMNS_8_ANA\FY_KE_DUCT_OSF_SENSITIVITY_PUSHOVER_MDOF+

21 %



IPython Console | Plots | Variable Explorer | Help | Debugger | History | Files

Inline | Conda: anaconda3 (Python 3.12.7) | LSP: Python | Line 1, Col 1 | UTF-8 | CRLF | RW | Mem 69%



$$\Delta_d = \frac{\sum_{i=1}^{n} m_i \Delta_i^2}{\sum_{i=1}^{n} m_i \Delta_i}$$

$$m_{eff} = \frac{\sum_{i=1}^{n} m_i \Delta_i}{\Delta_d}$$

$$k_{eff} = \frac{\sum_{i=1}^{n} k_i \Delta_i}{\Delta_d}$$

$$T_{eff} = \frac{2\pi}{\sqrt{\frac{k_{eff}}{m_{eff}}}}$$

BILINEAR CURVE

```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
---------------------------
[[0.00000000e+00 0.00000000e+00]
 [1.05586496e-01 3.45555806e+05]
 [9.44000000e-01 3.86740624e+05]]
+==========================+
+-------------------------------------------------+
 Structure Elastic Stiffness :    3272727.27
 Structure Plastic Stiffness :    409682.86
 Structure Tangent Stiffness :    [3272727.27   49122.32]
 Structure Ductility Ratio :      8.94
 Structure Over Strength Factor:   1.12
+-------------------------------------------------+
```

TRILINEAR CURVE

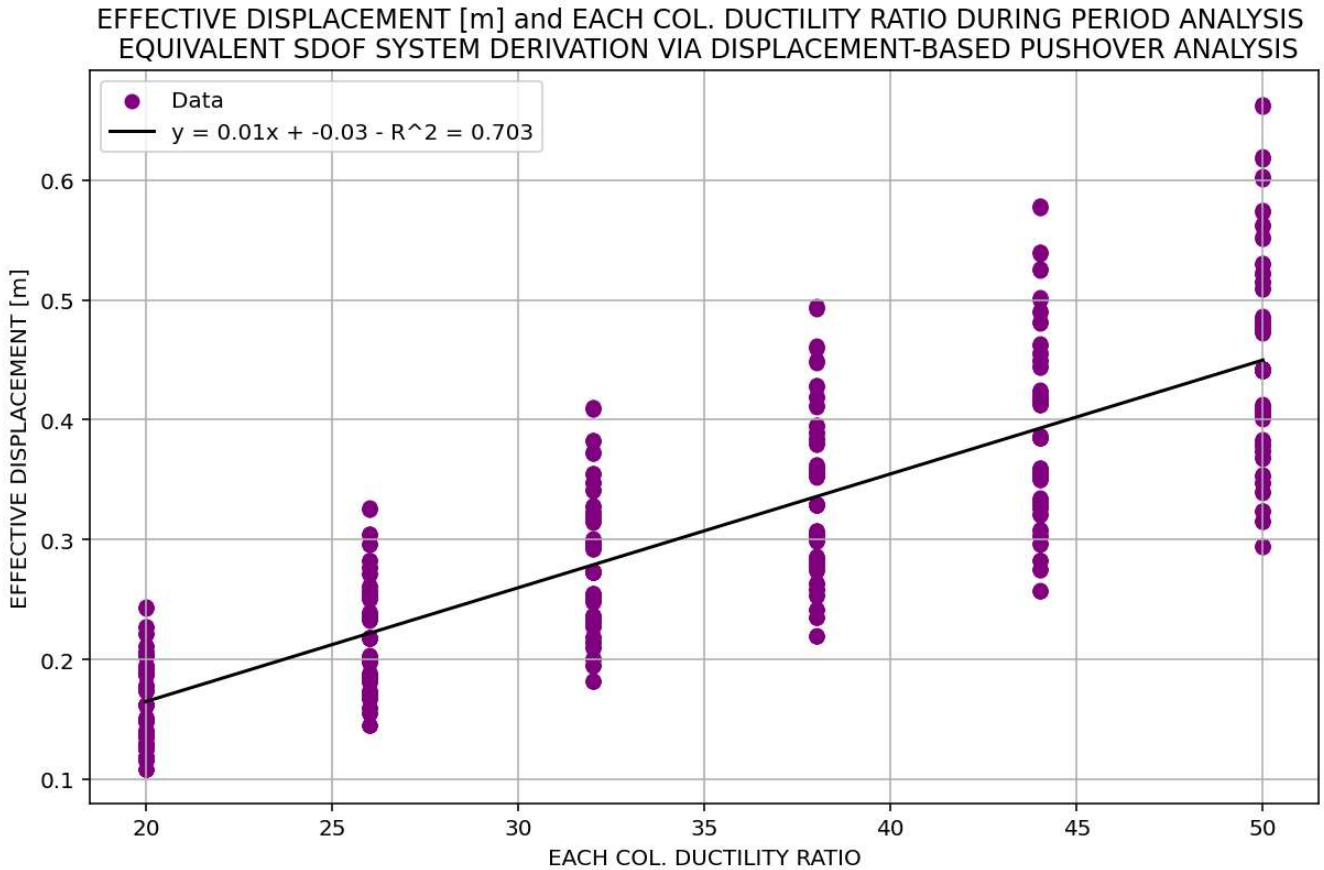
```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
---------------------------
[[0.00000000e+00 0.00000000e+00]
 [1.04512702e-01 3.42041571e+05]
 [5.67000000e-01 3.70124071e+05]
 [9.44000000e-01 3.86740624e+05]]
+==========================+
+-------------------------------------------------+
 Structure Elastic Stiffness :    3272727.27
 Structure Plastic Stiffness :    409682.86
 Structure Tangent Stiffness :    [3272727.27   60720.59   44075.74]
 Structure Ductility Ratio :      9.03
 Structure Over Strength Factor:   1.13
+-------------------------------------------------+
```

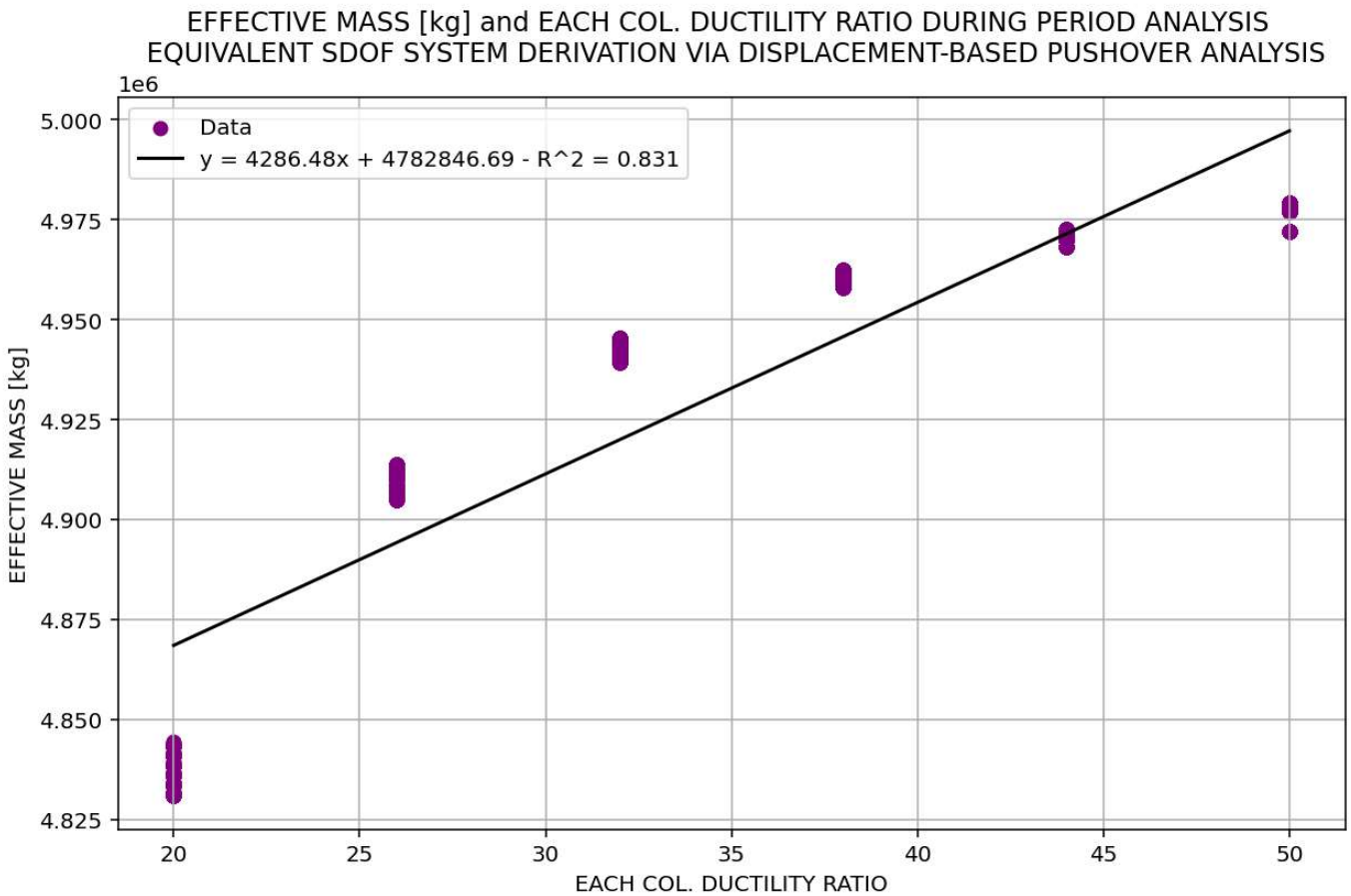
QUADRILINEAR CURVE

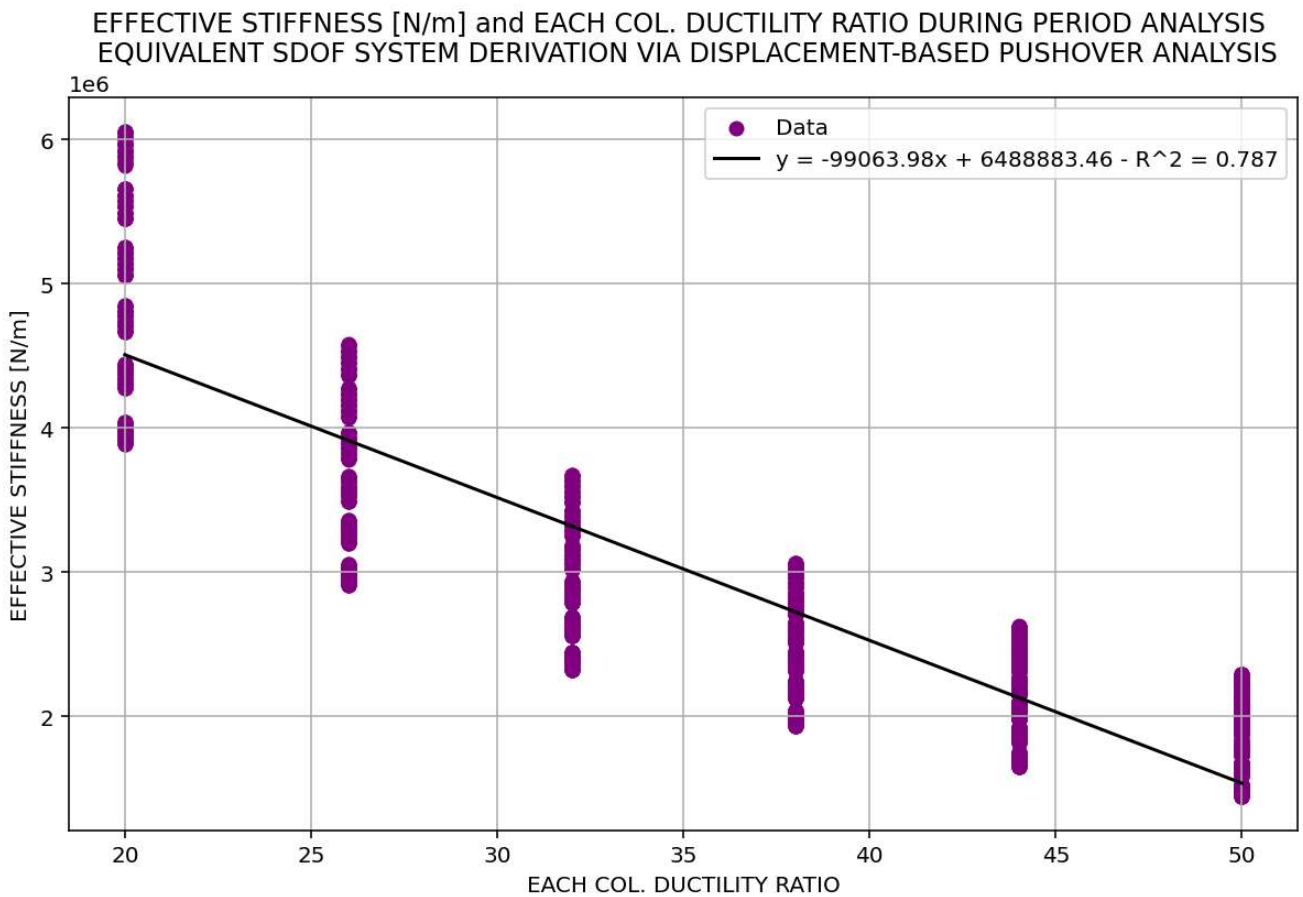
```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
---------------------------
[[0.00000000e+00 0.00000000e+00]
 [1.04262557e-01 3.41222915e+05]
 [5.67000000e-01 3.70124071e+05]
 [6.14000000e-01 3.73181289e+05]
 [9.44000000e-01 3.86740624e+05]]
+==========================+
+-------------------------------------------------+
 Structure Elastic Stiffness :    3272727.27
 Structure Plastic Stiffness :    409682.86
 Structure Tangent Stiffness :    [3272727.27   62456.92   65047.18   41088.89]
 Structure Ductility Ratio :      9.05
 Structure Over Strength Factor:   1.13
+-------------------------------------------------+
```

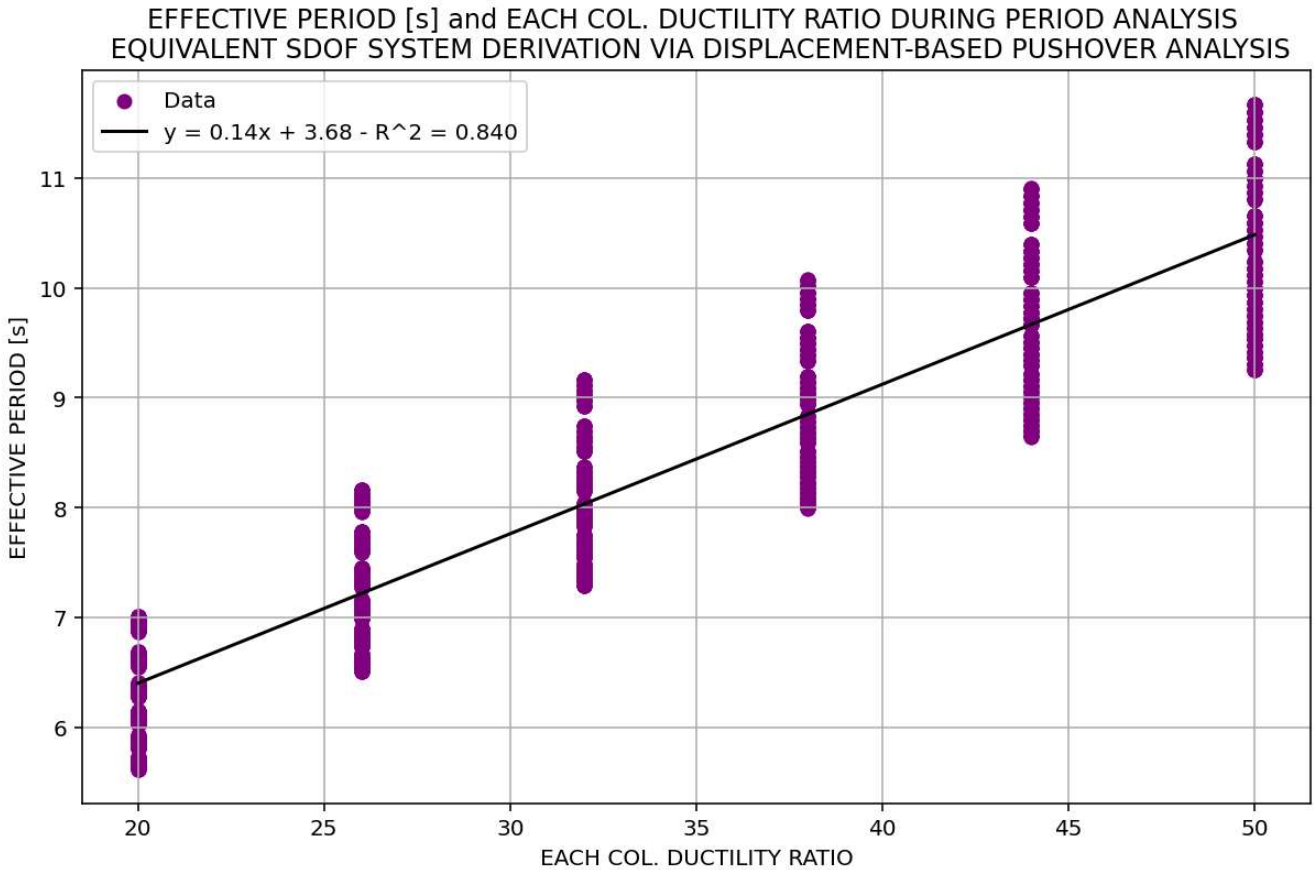
PENTALINEAR CURVE

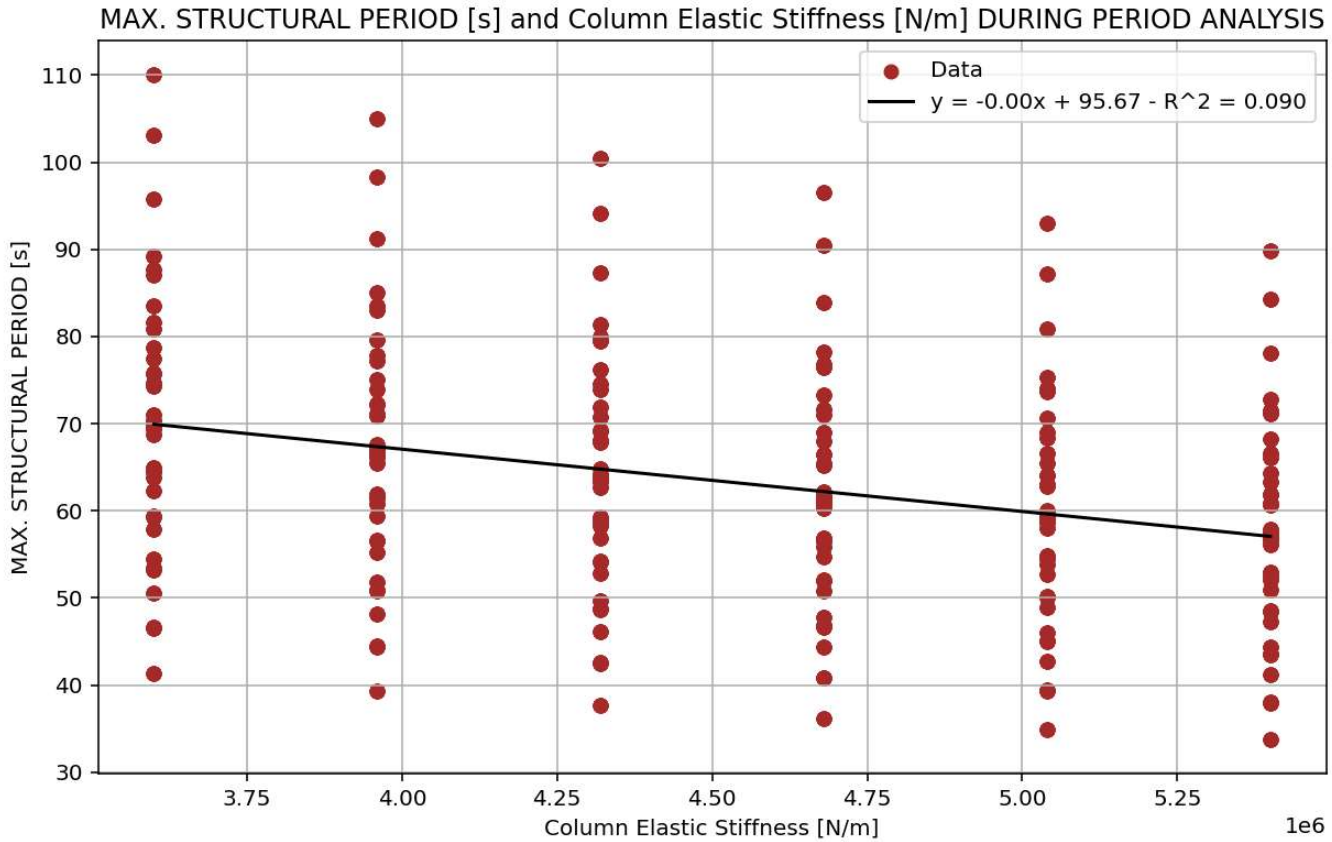
```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
---------------------------
[[0.00000000e+00 0.00000000e+00]
 [1.04126239e-01 3.40776781e+05]
 [5.67000000e-01 3.70124071e+05]
 [6.14000000e-01 3.73181289e+05]
 [8.50000000e-01 3.83491914e+05]
 [9.44000000e-01 3.86740624e+05]]
+==========================+
+-------------------------------------------------+
 Structure Elastic Stiffness :    3272727.27
 Structure Plastic Stiffness :    409682.86
 Structure Tangent Stiffness :    [3272727.27   63402.36   65047.18   43689.09   34560.74]
 Structure Ductility Ratio :      9.07
 Structure Over Strength Factor:   1.13
+-------------------------------------------------+
```

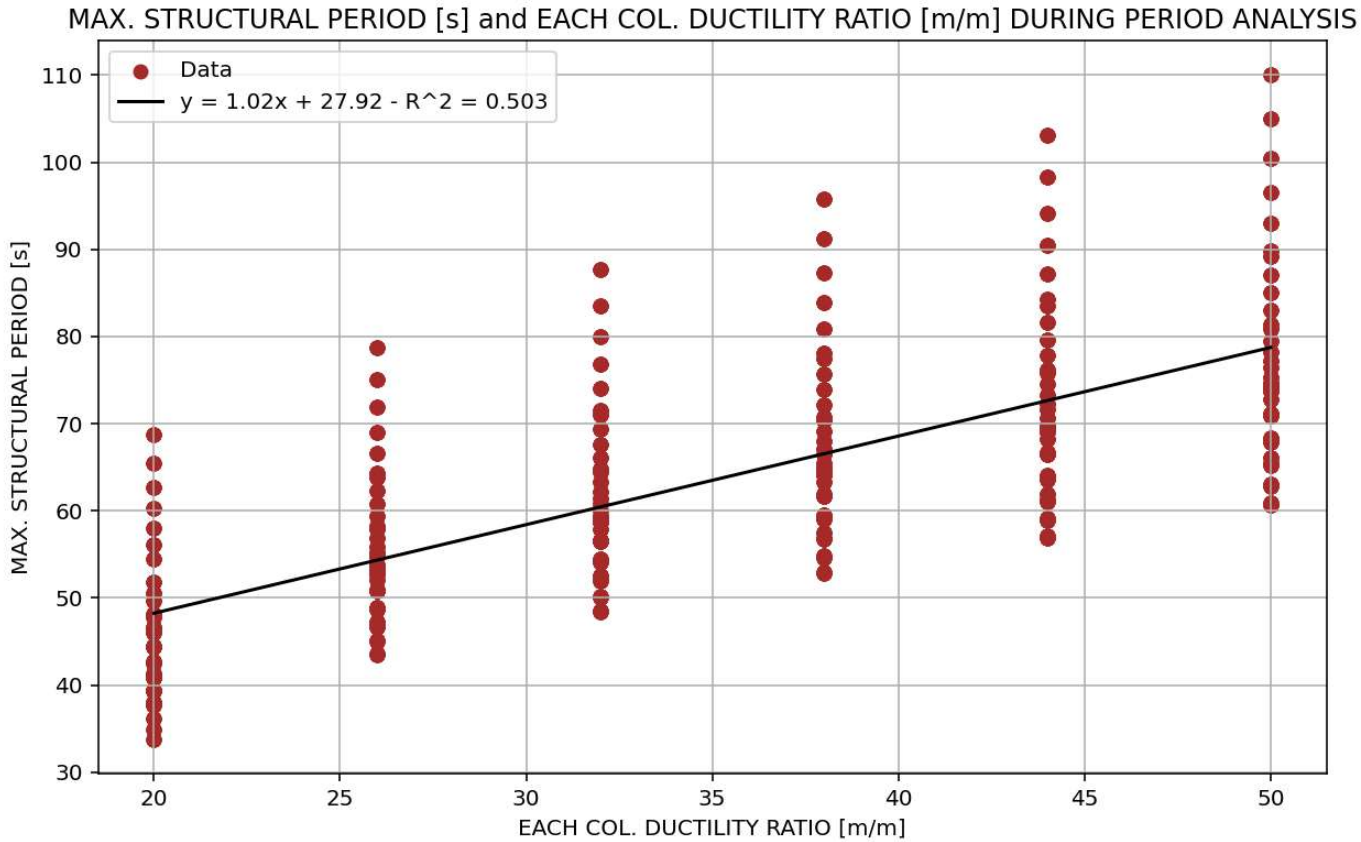

EFFECTIVE DISPLACEMENT [m] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS
Data
y = 0.01x + -0.03 - R^2 = 0.703
EFFECTIVE DISPLACEMENT [m]
0.1
0.2
0.3
0.4
0.5
0.6
20
25
30
35
40
45
50
EACH COL. DUCTILITY RATIO


EFFECTIVE MASS [kg] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS
1e6
Data
y = 4286.48x + 4782846.69 - R^2 = 0.831
5.000
4.975
4.950
4.925
4.900
4.875
4.850
4.825
EFFECTIVE MASS [kg]
20
25
30
35
40
45
50
EACH COL. DUCTILITY RATIO

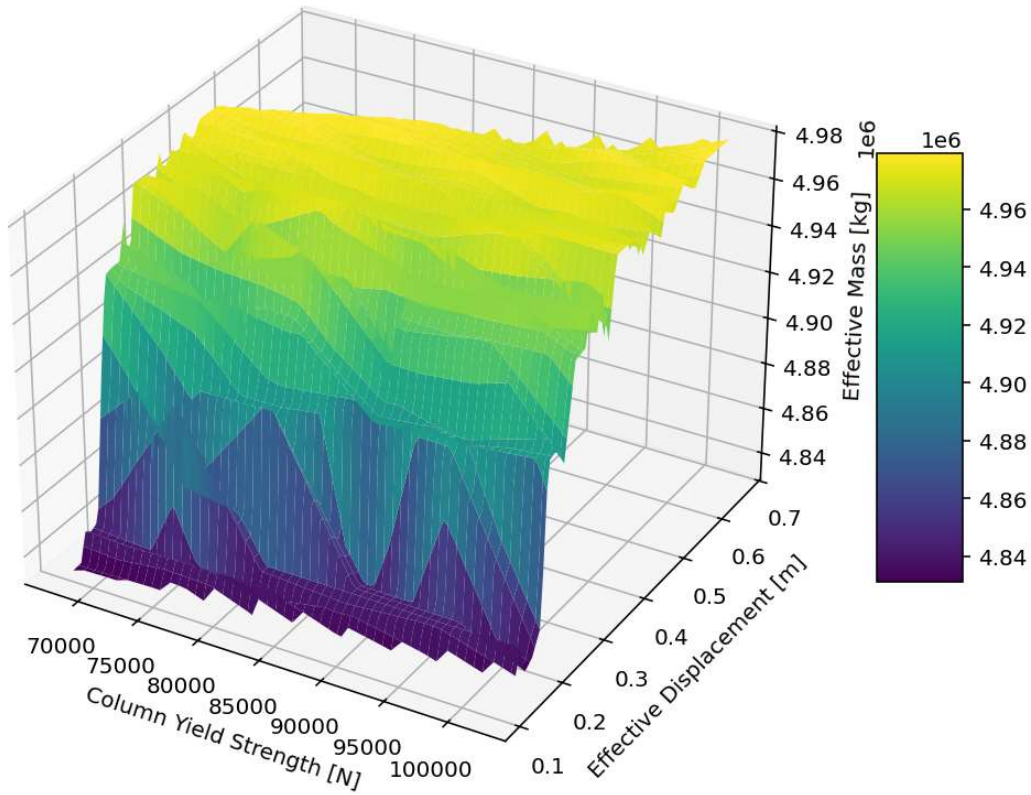

EFFECTIVE STIFFNESS [N/m] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS
1e6
Data
y = -99063.98x + 6488883.46 - R^2 = 0.787
EFFECTIVE STIFFNESS [N/m]
EACH COL. DUCTILITY RATIO
6
5
4
3
2
20
25
30
35
40
45
50


EFFECTIVE PERIOD [s] and EACH COL. DUCTILITY RATIO DURING PERIOD ANALYSIS
EQUIVALENT SDOF SYSTEM DERIVATION VIA DISPLACEMENT-BASED PUSHOVER ANALYSIS
Data
y = 0.14x + 3.68 - R^2 = 0.840
EFFECTIVE PERIOD [s]
EACH COL. DUCTILITY RATIO
6
7
8
9
10
11
20
25
30
35
40
45
50

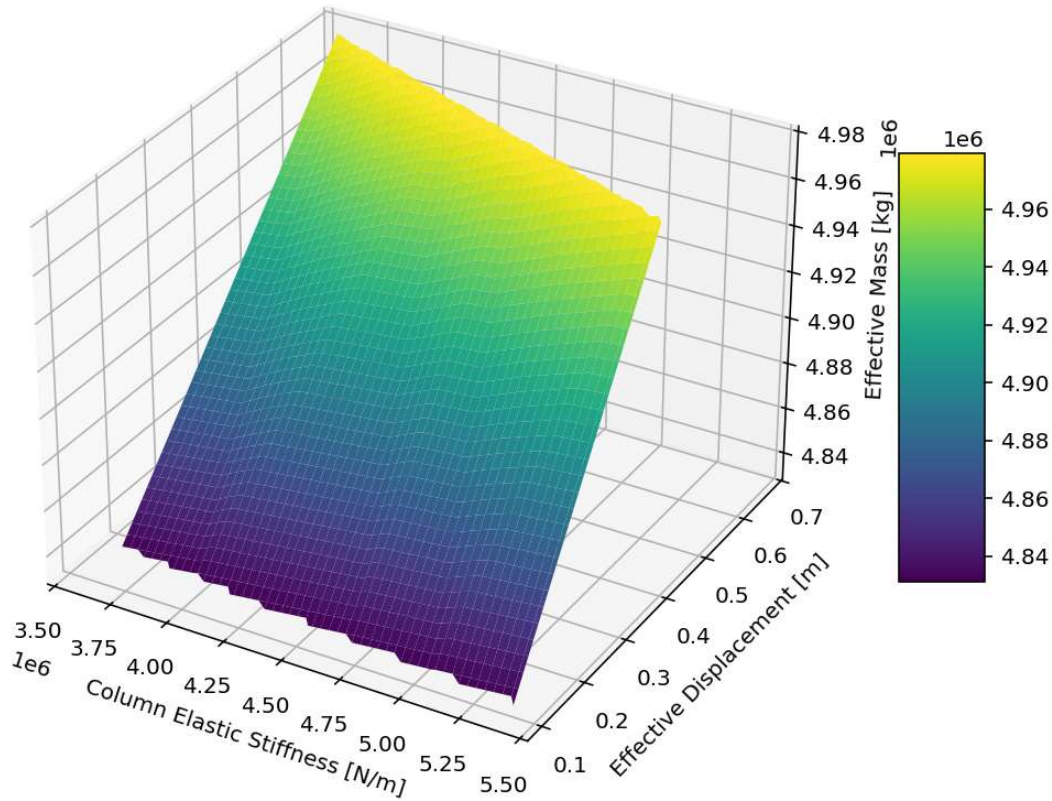

MAX. STRUCTURAL PERIOD [s] and Column Elastic Stiffness [N/m] DURING PERIOD ANALYSIS
Data
y = -0.00x + 95.67 - R^2 = 0.090
MAX. STRUCTURAL PERIOD [s]
110
100
90
80
70
60
50
40
30
3.75
4.00
4.25
4.50
4.75
5.00
5.25
Column Elastic Stiffness [N/m]
1e6


MAX. STRUCTURAL PERIOD [s] and EACH COL. DUCTILITY RATIO [m/m] DURING PERIOD ANALYSIS
Data
y = 1.02x + 27.92 - R^2 = 0.503
MAX. STRUCTURAL PERIOD [s]
EACH COL. DUCTILITY RATIO [m/m]
110
100
90
80
70
60
50
40
30
20
25
30
35
40
45
50

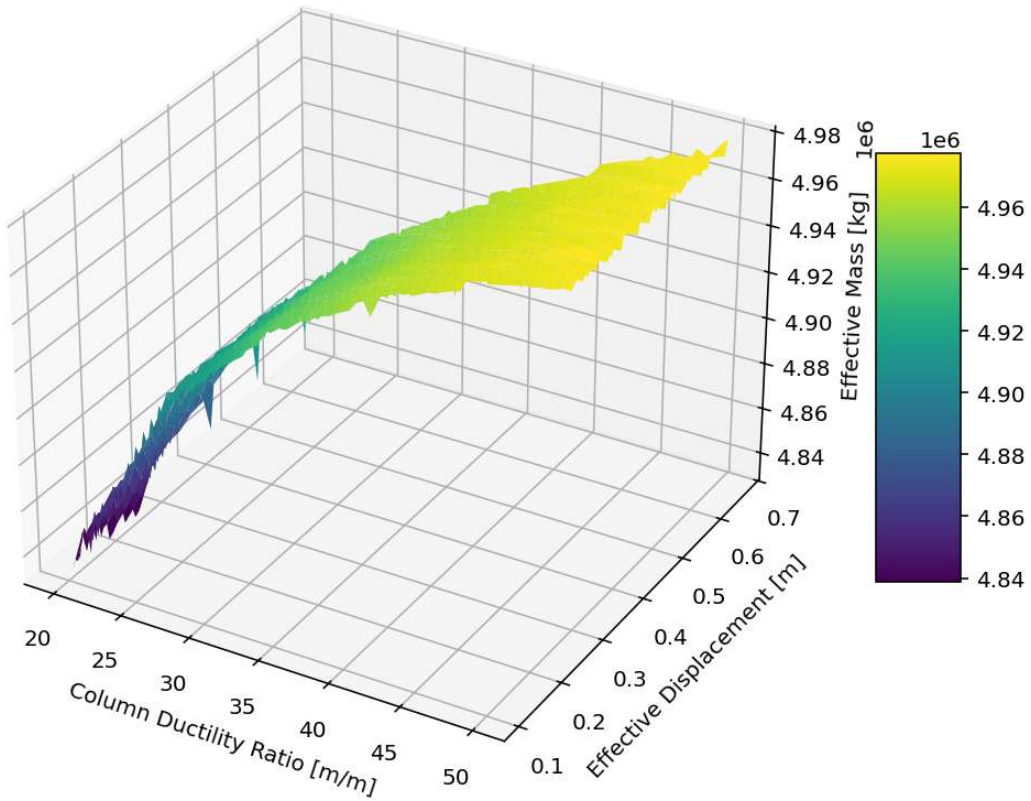
3D CONTOUR: Column Yield Strength [N] - Effective Displacement [m] - Effective Mass [kg]



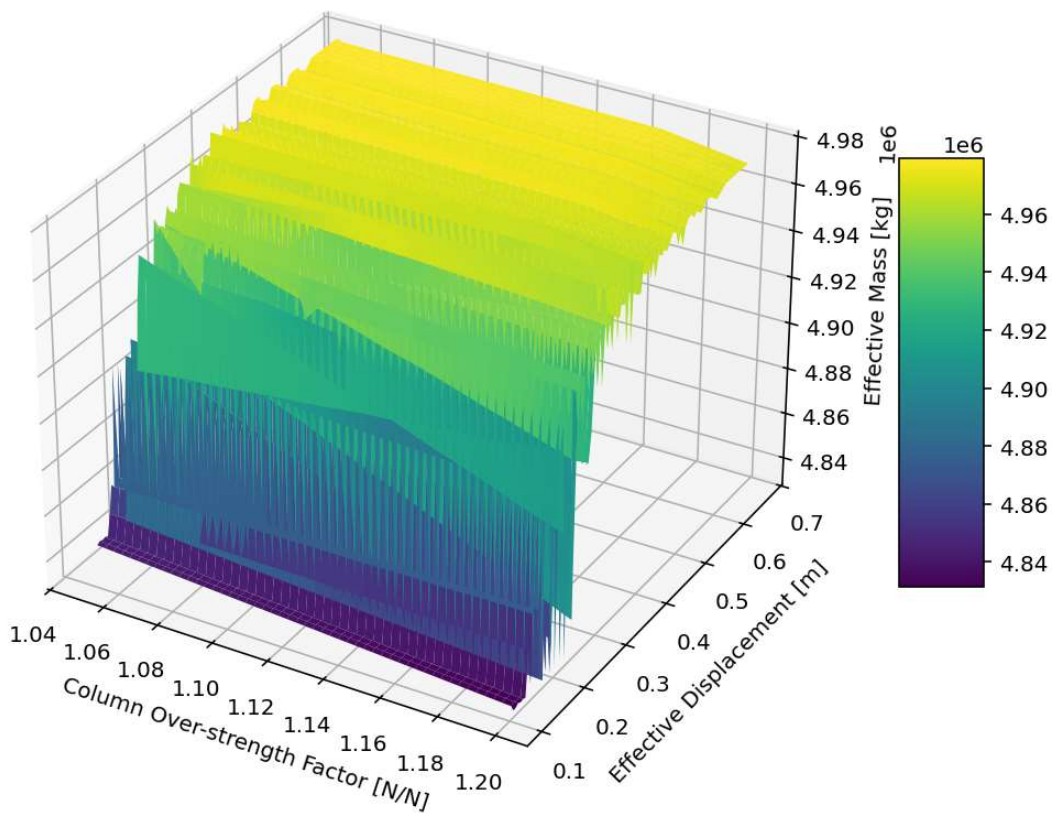
3D CONTOUR: Column Elastic Stiffness [N/m] - Effective Displacement [m] - Effective Mass [kg]



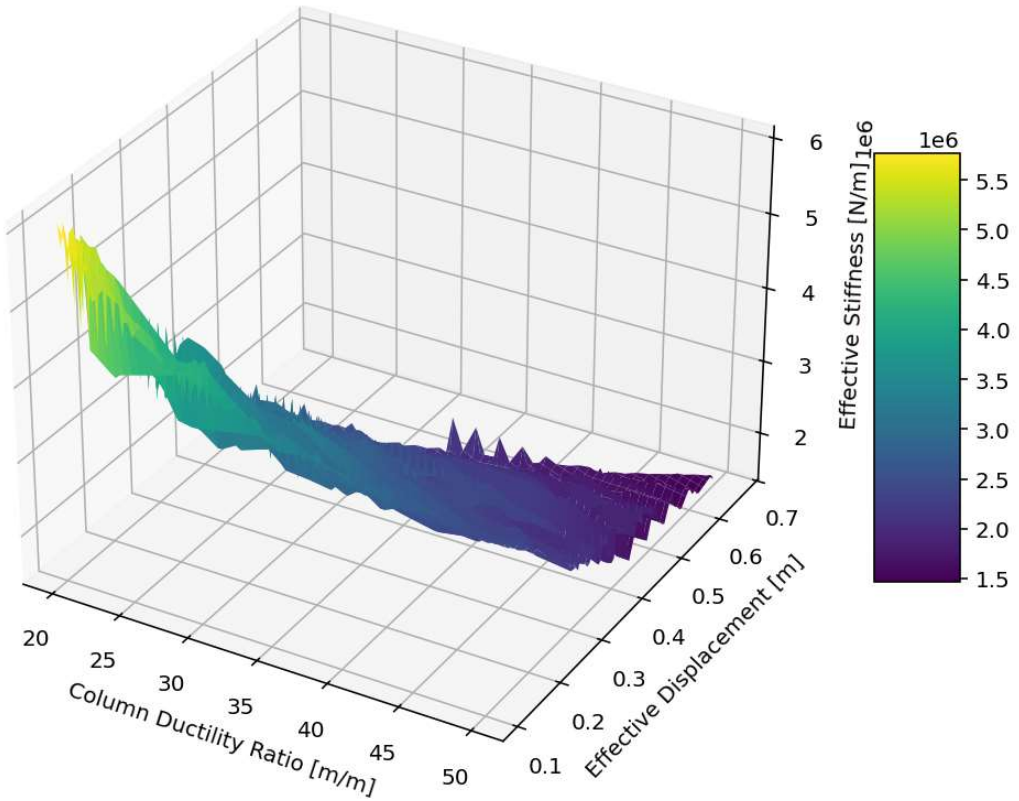
3D CONTOUR: Column Ductility Ratio [m/m] - Effective Displacement [m] - Effective Mass [kg]



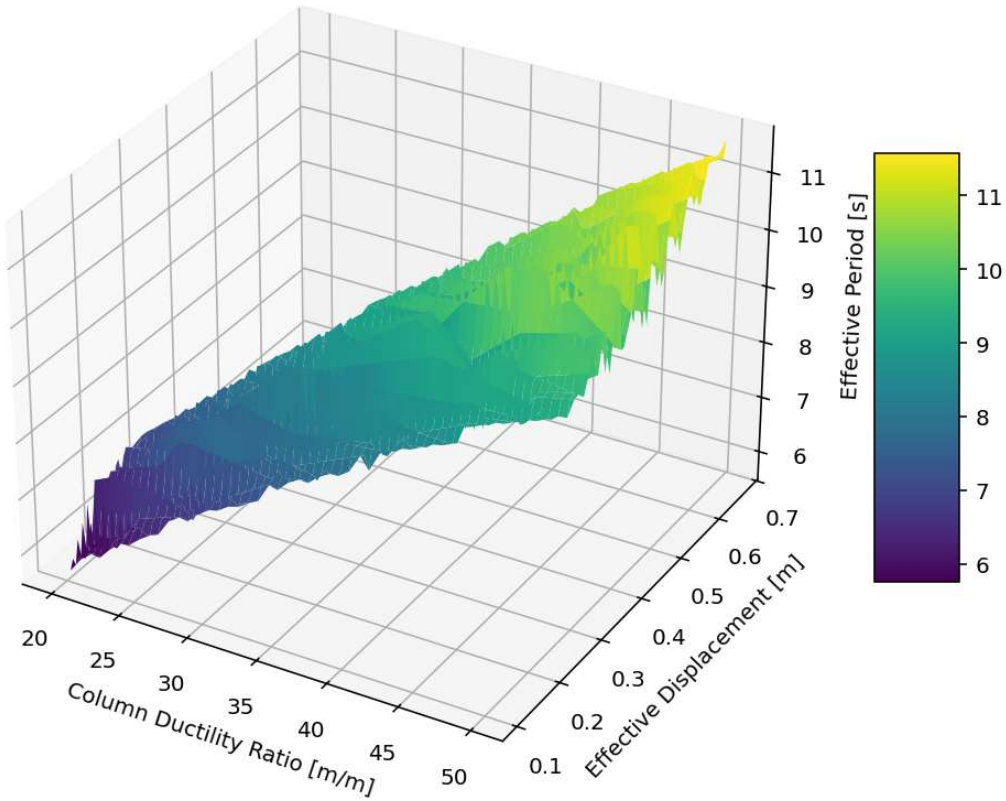
3D CONTOUR: Column Over-strength Factor [N/N] - Effective Displacement [m] - Effective Mass [kg]

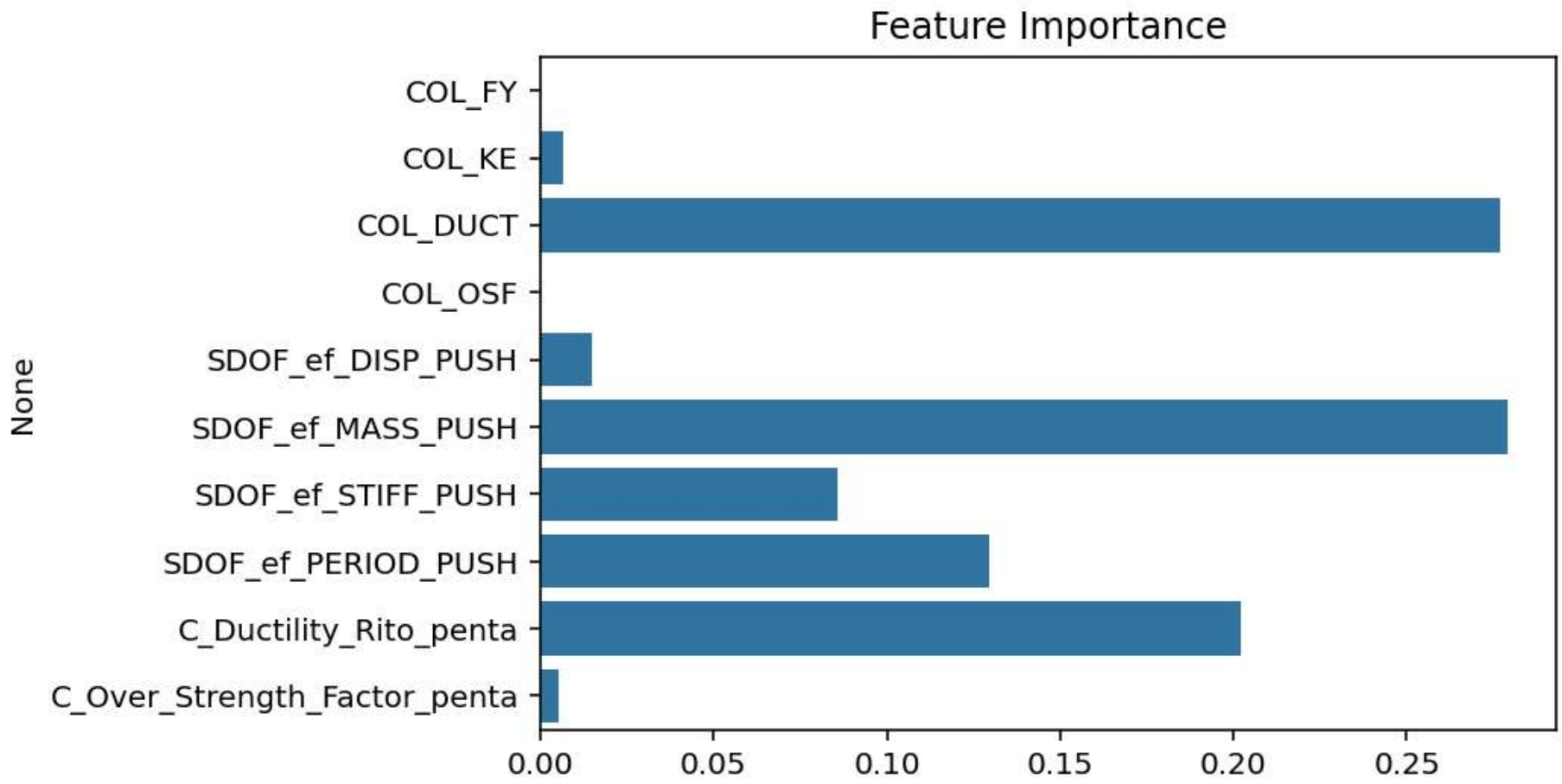


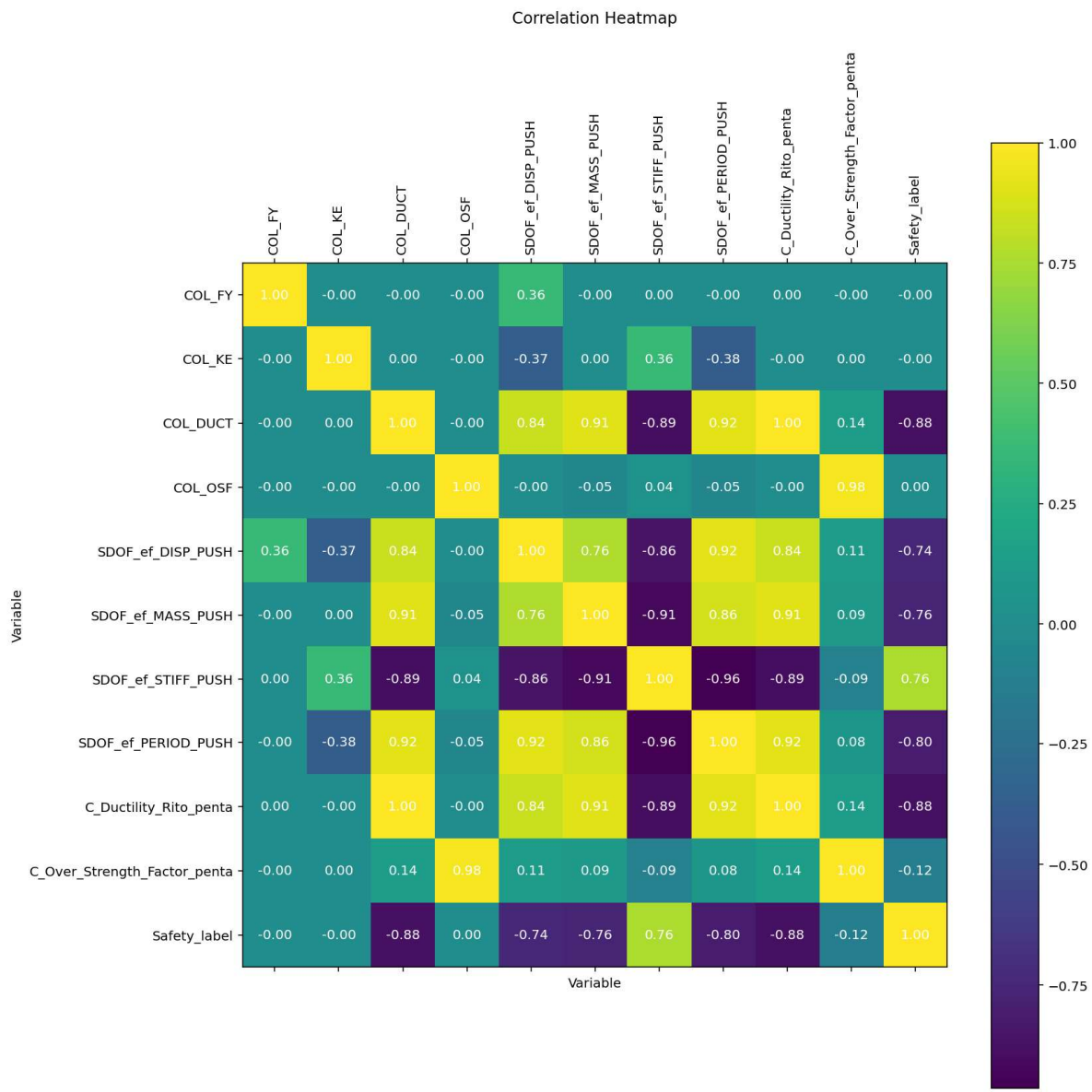
3D CONTOUR: Column Ductility Ratio [m/m] - Effective Displacement [m] - Effective Stiffness [N/m]



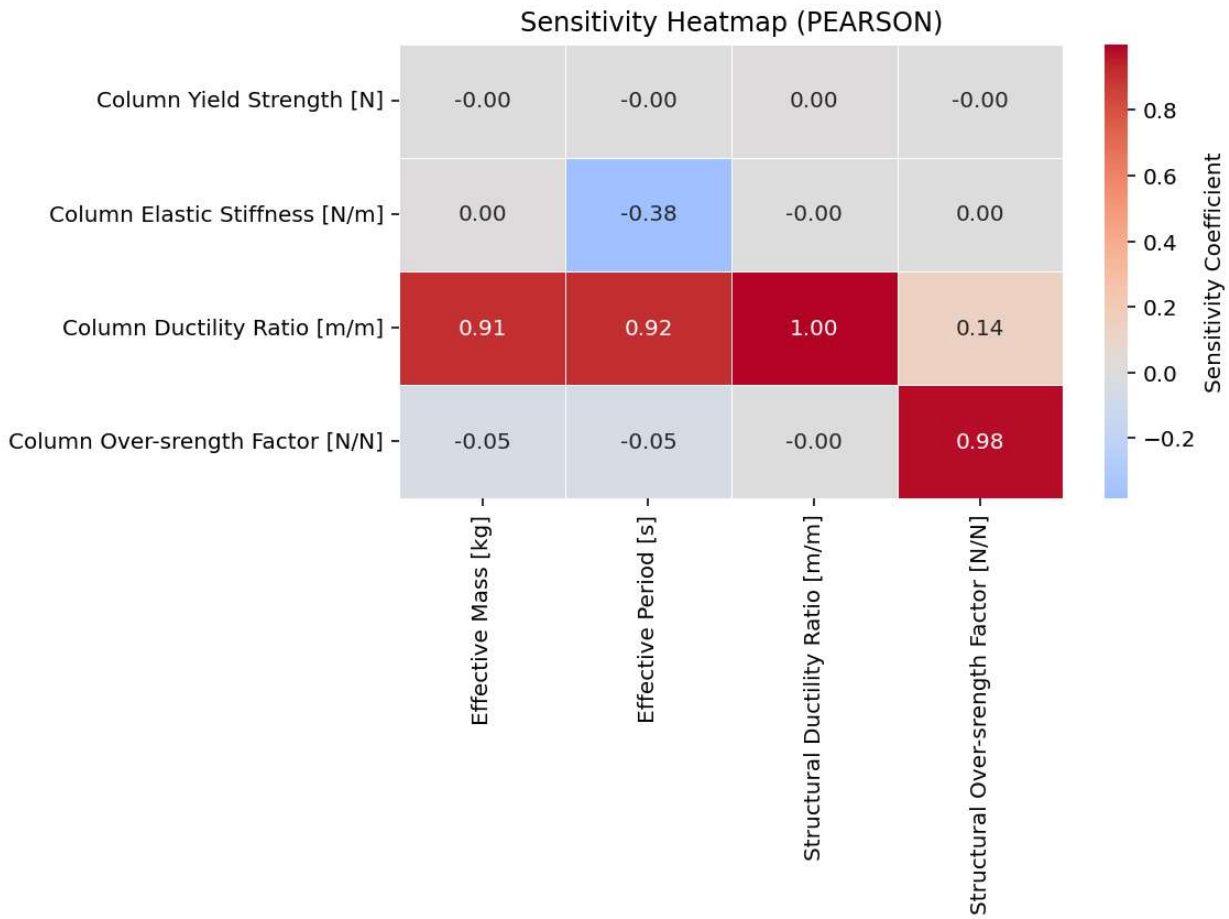
3D CONTOUR: Column Ductility Ratio [m/m] - Effective Displacement [m] - Effective Period [s]

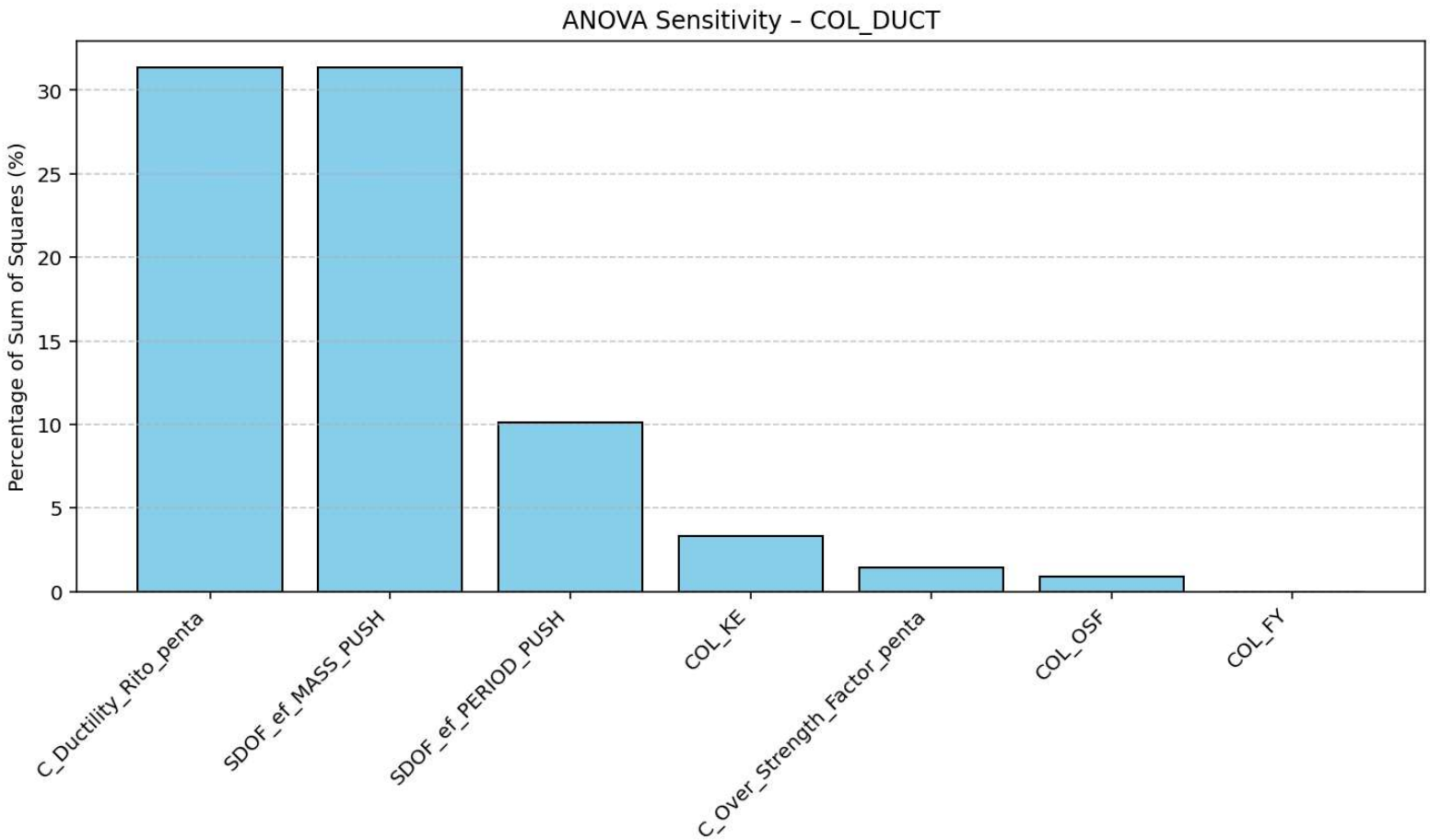


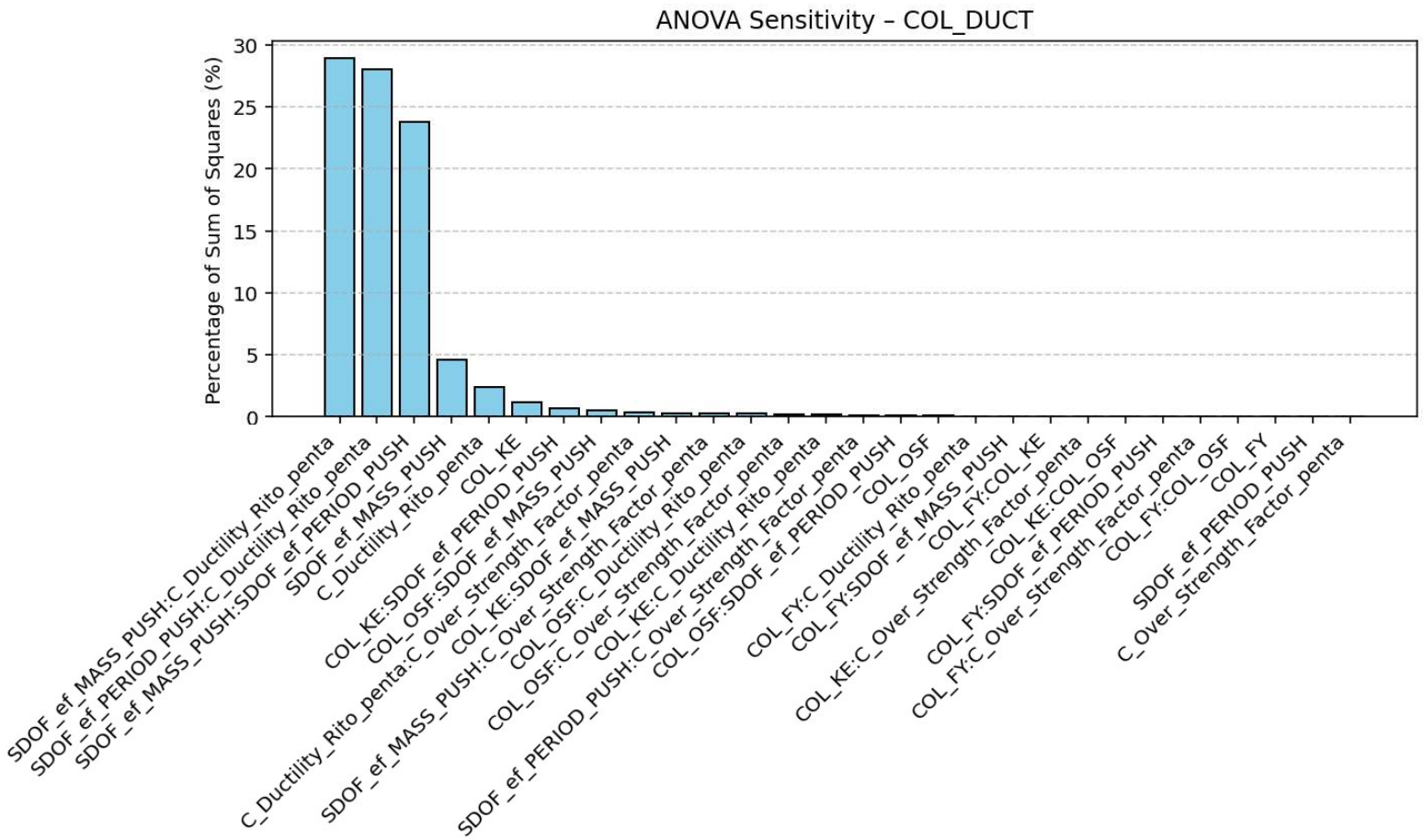

Feature Importance
None
COL_FY
COL_KE
COL_DUCT
COL_OSF
SDOF_ef_DISP_PUSH
SDOF_ef_MASS_PUSH
SDOF_ef_STIFF_PUSH
SDOF_ef_PERIOD_PUSH
C_Ductility_Rito_penta
C_Over_Strength_Factor_penta
0.00
0.05
0.10
0.15
0.20
0.25

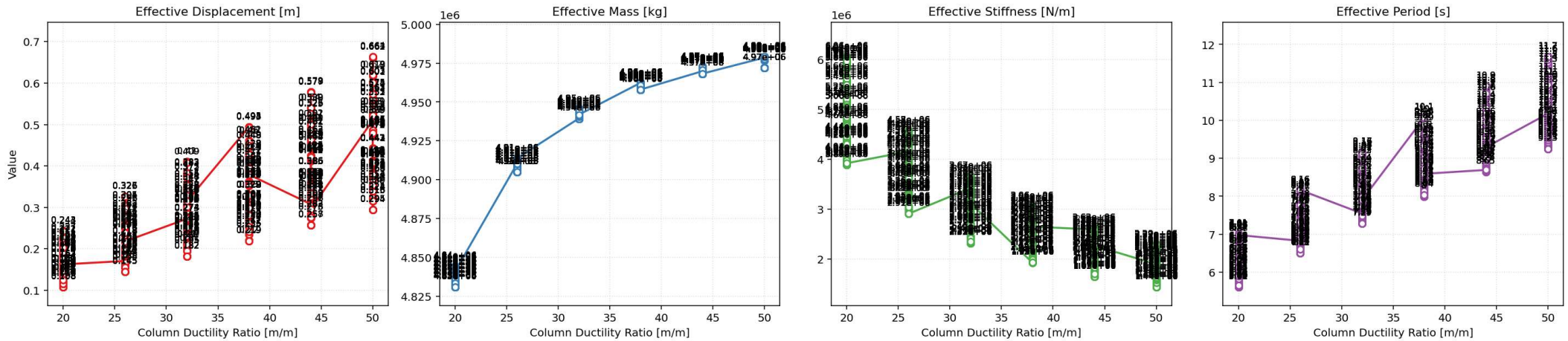


Variable	COL_FY	COL_KE	COL_DUCT	COL_OSF	SDOF_ef_DISP_PUSH	SDOF_ef_MASS_PUSH	SDOF_ef_STIFF_PUSH	SDOF_ef_PERIOD_PUSH	C_Ductility_Rito_penta	C_Over_Strength_Factor_penta	Safety_label
COL_FY	1.00	-0.00	-0.00	-0.00	0.36	-0.00	0.00	-0.00	0.00	-0.00	-0.00
COL_KE	-0.00	1.00	0.00	-0.00	-0.37	0.00	0.36	-0.38	-0.00	0.00	-0.00
COL_DUCT	-0.00	0.00	1.00	-0.00	0.84	0.91	-0.89	0.92	1.00	0.14	-0.88
COL_OSF	-0.00	-0.00	-0.00	1.00	-0.00	-0.05	0.04	-0.05	-0.00	0.98	0.00
SDOF_ef_DISP_PUSH	0.36	-0.37	0.84	-0.00	1.00	0.76	-0.86	0.92	0.84	0.11	-0.74
SDOF_ef_MASS_PUSH	-0.00	0.00	0.91	-0.05	0.76	1.00	-0.91	0.86	0.91	0.09	-0.76
SDOF_ef_STIFF_PUSH	0.00	0.36	-0.89	0.04	-0.86	-0.91	1.00	-0.96	-0.89	-0.09	0.76
SDOF_ef_PERIOD_PUSH	-0.00	-0.38	0.92	-0.05	0.92	0.86	-0.96	1.00	0.92	0.08	-0.80
C_Ductility_Rito_penta	0.00	-0.00	1.00	-0.00	0.84	0.91	-0.89	0.92	1.00	0.14	-0.88
C_Over_Strength_Factor_penta	-0.00	0.00	0.14	0.98	0.11	0.09	-0.09	0.08	0.14	1.00	-0.12
Safety_label	-0.00	-0.00	-0.88	0.00	-0.74	-0.76	0.76	-0.80	-0.88	-0.12	1.00


Sensitivity Heatmap (PEARSON)
Column Yield Strength [N]
Column Elastic Stiffness [N/m]
Column Ductility Ratio [m/m]
Column Over-srength Factor [N/N]
-0.00
-0.00
0.00
-0.00
0.00
-0.38
-0.00
0.00
0.91
0.92
1.00
0.14
-0.05
-0.05
-0.00
0.98
Effective Mass [kg]
Effective Period [s]
Structural Ductility Ratio [m/m]
Structural Over-srength Factor [N/N]
0.8
0.6
0.4
0.2
0.0
−0.2
Sensitivity Coefficient


ANOVA Sensitivity – COL_DUCT
Percentage of Sum of Squares (%)
0
5
10
15
20
25
30
C_Ductility_Rito_penta
SDOF_ef_MASS_PUSH
SDOF_ef_PERIOD_PUSH
COL_KE
C_Over_Strength_Factor_penta
COL_OSF
COL_FY


ANOVA Sensitivity – COL_DUCT
Percentage of Sum of Squares (%)
0
5
10
15
20
25
30
SDOF_ef_MASS_PUSH:C_Ductility_Rito_penta
SDOF_ef_PERIOD_PUSH:C_Ductility_Rito_penta
SDOF_ef_MASS_PUSH:SDOF_ef_PERIOD_PUSH
SDOF_ef_MASS_PUSH
C_Ductility_Rito_penta
COL_KE
COL_KE:SDOF_ef_PERIOD_PUSH
COL_OSF:SDOF_ef_MASS_PUSH
C_Ductility_Rito_penta:C_Over_Strength_Factor_penta
COL_KE:SDOF_ef_MASS_PUSH
SDOF_ef_MASS_PUSH:C_Over_Strength_Factor_penta
COL_OSF:C_Over_Strength_Factor_penta
COL_OSF:C_Ductility_Rito_penta
SDOF_ef_PERIOD_PUSH:C_Over_Strength_Factor_penta
COL_KE:C_Ductility_Rito_penta
COL_OSF:SDOF_ef_PERIOD_PUSH
COL_OSF
COL_FY:C_Ductility_Rito_penta
COL_FY:SDOF_ef_MASS_PUSH
COL_FY:COL_KE
COL_KE:C_Over_Strength_Factor_penta
COL_KE:COL_OSF
COL_FY:SDOF_ef_PERIOD_PUSH
COL_FY:C_Over_Strength_Factor_penta
COL_FY:COL_OSF
COL_FY
SDOF_ef_PERIOD_PUSH
C_Over_Strength_Factor_penta


Effective Displacement [m]
Value
Column Ductility Ratio [m/m]
Effective Mass [kg]
1e6
Column Ductility Ratio [m/m]
Effective Stiffness [N/m]
1e6
Column Ductility Ratio [m/m]
Effective Period [s]
Column Ductility Ratio [m/m]